

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

MYLAN PHARMACEUTICALS, INC., MSN LABORATORIES
PRIVATE LTD. and MSN PHARMACEUTICALS INC.,
Petitioners,

v.

BAUSCH HEALTH IRELAND LIMITED,
Patent Owner.

IPR2022-00722¹
Patent 7,041,786 B2

Before TINA E. HULSE, CYNTHIA M. HARDMAN, and
MICHAEL A. VALEK, *Administrative Patent Judges*.

VALEK, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)
Granting-in-Part Patent Owner's Motion to Exclude
Denying Petitioner's Motion to Exclude
37 C.F.R. § 42.64(c)

¹ IPR2023-00016 has been joined with this proceeding.

I. INTRODUCTION

Mylan Pharmaceuticals Inc. (“Mylan”) filed a Petition (Paper 1, “Pet.”), seeking *inter partes* review of claims 1–6 of U.S. Patent No. 7,041,786 B2 (Ex. 1001, “the ’786 patent”). We instituted trial on all of the grounds in the Petition. Paper 16, 32.

Following institution, MSN Laboratories Private Ltd. and MSN Pharmaceuticals Inc. (collectively “MSN”) filed a substantially identical petition in IPR2023-00016. That proceeding was instituted and joined with this one. See Paper 25 (granting joinder). Herein, we refer to Mylan and MSN collectively as “Petitioner.”²

Bausch Health Ireland Limited (“Patent Owner”) then filed a Response (Paper 70,³ “Resp.”), Petitioner filed a Reply (Paper 38, “Reply”), and Patent Owner filed a Sur-reply (Paper 71, “Sur-reply”). We held a hearing on June 14, 2023, and a transcript is of record. Paper 75 (“Tr.”). In addition, both Parties have filed motions to exclude certain exhibits. Paper 53 (“PO MTE”); Paper 67 (“Pet. MTE”).

After considering the parties’ arguments and evidence, we find that Petitioner has not shown by a preponderance of the evidence that the challenged claims of the ’786 patent are unpatentable. See 35 U.S.C. § 316(e). Moreover, as explained below, we grant Patent Owner’s motion to exclude in part and deny Petitioner’s motion to exclude.

² MSN has agreed to an “understudy role” in this proceeding. See IPR2023-00016, Paper 14.

³ Portions of Patent Owner’s Response and Sur-reply were initially filed under seal. See Papers 28 and 51. Patent Owner has since filed unredacted versions that we refer to herein. See Paper 73 (Patent Owner agreeing to make these papers public).

II. BACKGROUND

A. Real Parties in Interest

In addition to itself, Mylan identifies the following real parties in interest (“RPI”): Mylan Inc. and Viatris Inc. *See* Paper 9, 2 (identifying these entities as parent companies); Paper 16, 31 (accepting Petitioner’s representation that these entities are RPI). MSN identifies MSN Laboratories Private Limited and MSN Pharmaceuticals, Inc. as RPI. IPR2023-00016, Paper 16, 2. Patent Owner identifies itself and Salix Pharmaceuticals, Inc. as RPI. Paper 4, 2.

B. The ’786 Patent

The ’786 patent issued on May 9, 2006, from a utility application filed on March 28, 2002, and claims priority to a provisional application filed on January 17, 2002. Ex. 1001, codes (22), (45), (60).

The ’786 patent is titled “Guanylate Cyclase Receptor Agonists for the Treatment of Tissue Inflammation and Carcinogenesis.” Ex. 1001, code (54). According to the Specification,

[t]he present invention relates to the therapeutic use of guanylate cyclase receptor agonists as a means for enhancing the intracellular production of cGMP. The agonists may be used either alone or in combination with inhibitors of cGMP-specific phosphodiesterase to prevent or treat cancerous, pre-cancerous and metastatic growths, particularly in the gastrointestinal tract and lungs. In addition, the agonists may be used in the treatment of inflammatory disorders such as ulcerative colitis and asthma.

Id. at 1:14–23.

The Specification explains that “[u]roguanylin, guanylin and bacterial ST [i.e., heat-stable enterotoxin] peptides are structurally related peptides

that bind to a guanylate cyclase receptor and stimulate intracellular production of cyclic guanosine monophosphate (cGMP).” Ex. 1001, 1:26–30. “This results in the activation of the cystic fibrosis transmembrane conductance regulator (CFTR),” which in turn “leads to stimulation of sodium and water secretion into the intestinal lumen.” *Id.* at 1:30–36. “Therefore, by serving as paracrine regulators of CFTR activity, cGMP receptor agonists,” such as uroguanylin, guanylin, and bacterial ST peptides, “regulate fluid and electrolyte transport in the GI tract.” *Id.* at 1:36–39.

“The present invention is based upon the development of new agonists of guanylate cyclase receptor” that “are analogs of uroguanylin.” Ex. 1001, 3:3–6. The Specification describes the sequence for its “most preferred” uroguanylin analog in SEQ ID NO:20, which is reproduced below.

Asn¹ Asp² Glu³ Cys⁴ Glu⁵ Leu⁶ Cys⁷ Val⁸ Asn⁹ Val¹⁰ Ala¹¹ Cys¹² Thr¹³ Gly¹⁴ Cys¹⁵ Leu¹⁶

Id. at 5:7–12. SEQ ID NO:20 provides the primary structure of a 16-amino acid long peptide. The sequence in SEQ ID NO:20 differs from that of the “Parent compound: uroguanylin” (shown in SEQ ID NO: 1) in that it has a “Glu³,” i.e., a glutamic acid (Glu) at position 3 in the peptide sequence, whereas human uroguanylin has an “Asp³,” i.e., an aspartic acid (Asp) at that position.⁴ *See id.* at 11 (Table 2). This is the only difference between the

⁴ Both parties use a superscript to denote the position of the residue in an amino acid sequence. *See, e.g.*, Pet. 1, 38; Resp. 2 (referring to “Glu³” and “Asp³”). In addition to Glu, other synonyms for glutamic acid appearing in the record include “glutamate” and the letter “E” from the single-letter code for amino acids. In addition to Asp, other synonyms for aspartic acid include “aspartate” and the letter “D.”

sequence in SEQ ID NO:20 and the naturally-occurring human uroguanylin sequence in SEQ ID NO:1. *Id.*; see also Ex. 1002 ¶¶ 24–25.

According to the Specification, the peptide of SEQ ID NO:20 has been found to have enhanced biological activity as an agonist of cGMP production due to its enhanced binding constant for the guanylate cyclase receptor, and is superior to uroguanylin with regard to temperature and protease stability and with regard to its biological activity at the physiologically favorable pH range (pH 6 to 7) in the large intestine.

Ex. 1001, 5:16–23. Table 4, reproduced below, shows data from a “T84 cell-based assay for determining the intracellular levels of cGMP.” *Id.* at 15:35–36.

TABLE 4

Peptide agonists evaluated for biological activity in the T84 cell bioassay.		
SEQ ID NO.*	Compound Code	cGMP Level** (pmol/well)
1	SP301	205
6	SP302	225
7	SP303	195
20	SP304	315
14	SP306	0
4	SP310	0
21	SP316	275

*SEQ ID's for SP301, SP304 and SP316 are the precise amino acid sequences for these analogs as given in the text.

**Intracellular cGMP level observed in T84 cells following treatment with 1 micromolar solution of the respective peptide agonist for 30 minutes. The value observed for SP304 was statistically significant with a $p > 0.5$.

“As shown in Table 4, SP304 (SEQ ID NO:20) gave the greatest enhancement of intracellular cGMP of all the” peptides tested. *Id.* at 15:53–58.

C. Challenged Claims

The Petition challenges claims 1–6. Claims 1–3 and 6 are independent claims. They provide as follows:

1. A peptide consisting of the amino acid sequence of SEQ ID NO:20.
2. A composition in unit dose comprising a guanylate cyclase receptor agonist peptide consisting of the amino acid sequence of SEQ ID NO:20.
3. A composition in unit dose form comprising:
 - a) a guanylate cyclase receptor agonist peptide consisting of the amino acid sequence of SEQ ID NO: 20; and
 - b) at least one compound selected from the group consisting of: a cGMP-dependent phosphodiesterase inhibitor, an anti-inflammatory agent, an antiviral agent and an anticancer agent.
6. A peptide conjugate comprising polyethylene glycol (PEG) attached to a peptide consisting of the amino acid sequence SEQ ID NO:20.

Ex. 1001, 37:1–12, 38:7–9. Claims 4 and 5 additionally limit the compositions of either claim 2 or 3 to a “dose form . . . selected from the group consisting of a tablet, a capsule, a solution and an inhalation formulation” (claim 4) and to further comprise “one or more excipients” (claim 5). *Id.* at 38:1–6.

Claim 1 is directed to a peptide with the sequence of SEQ ID NO:20. The parties’ papers and exhibits refer to this peptide by several names, including [Glu³]-human uroguanylin, [Glu³]-UG, plecanatide, and SP-304. *See, e.g.*, Pet. 5 (noting that Patent Owner “markets [Glu³]-human uroguanylin as TRULANCE[®] (plecanatide)”); Resp. 1; Ex. 1001, 10:42–63

(referring to the Glu³-substituted analog of human uroguanylin as SP304). Plecanatide “is the active pharmaceutical ingredient in Trulance®, which is FDA-approved for the treatment of chronic idiopathic constipation . . . and irritable bowel syndrome with constipation.” Resp. 1; *see also* Pet. 5. (similar); Ex. 1055 (prescribing information for “TRULANCE (plecanatide) tablets”).

D. Asserted Grounds of Unpatentability

Petitioners assert the following grounds of unpatentability:

Claim(s) Challenged	35 U.S.C. §⁵	Reference(s)/Basis
1	103(a)	Currie ⁶ , Li ⁷
2, 4, 5	103(a)	Currie, Li, Narayani ⁸
3–5	103(a)	Currie, Li, Narayani, Campieri ⁹
6	103(a)	Currie, Li, Ekwuribe ¹⁰

In support of these grounds, Petitioner relies on declarations from Blake R. Peterson, Ph.D. submitted with the Petition (Ex. 1002) and Reply (Ex. 1063) and a declaration from Michael S. Epstein, M.D. (Ex. 1064).

⁵ The Leahy-Smith America Invents Act, Pub. L. No. 112–29, 125 Stat. 284 (2011) (“AIA”), included revisions to 35 U.S.C. § 103 that became effective after the filing of the application that led to the ’786 patent. Therefore, we apply the pre-AIA version of 35 U.S.C. § 103.

⁶ U.S. Pat. 5,489,670, issued Feb. 6, 1996 (Ex. 1005) (“Currie”).

⁷ Z. Li et al., *Purification, cDNA Sequence, and Tissue Distribution of Rat Uroguanylin*, 68 *Regulatory Peptides* 45–56 (1997) (Ex. 1006) (“Li”).

⁸ R. Narayani et al., *Polymer-coated Gelatin Capsules as Oral Delivery Devices and their Gastrointestinal Tract Behavior in Humans*, 7 *J. Biomater. Sci. Polymer Edn.* 39–48 (1995) (Ex. 1007) (“Narayani”).

⁹ M. Campieri et al., *Oral Budesonide is as Effective as Oral Prednisolone in Active Crohn’s Disease*, 41 *Gut* 209–214 (1997) (Ex. 1008) (“Campieri”).

¹⁰ U.S. Pat. 5,359,030, issued Oct. 25, 1994 (Ex. 1009) (“Ekwuribe”).

Patent Owner relies on declarations from Kunwar Shailubhai (Ex. 2023, Ex. 2066), Stephen G. Davies (Ex. 2024), and Scott A. Waldman (Ex. 2025).

III. ANALYSIS OF THE ASSERTED GROUNDS

A. Legal Standards

Under pre-AIA § 103(a), a claim is unpatentable for obviousness if, to one of ordinary skill in the pertinent art, “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made.” 35 U.S.C. § 103(a) (2006); *see also KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) when in evidence, objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

B. Level of Ordinary Skill in the Art

Relying on the testimony of its declarant, Dr. Peterson, Petitioner contends that as of January 17, 2002:

A skilled artisan would have been familiar with signaling peptides and their biochemistry, as the accompanying exhibits prove. The specification admits an artisan would also have known how to choose and prepare various dosage forms when it notes the use of known alternatives without further guidance on when and how to use them. A Ph.D. in peptide chemistry, protein engineering, or a related field, or alternatively, a master’s degree in one of these fields plus two to five years of experience in drug development would represent typical education and experience for a skilled artisan. EX1002,

¶¶40-43. This individual would have worked in consultation with a team including, *e.g.*, a pharmaceutical chemist and/or a pharmacist familiar with formulating peptides for administration.

Pet. 13.

Patent Owner contends that a person of ordinary skill in the art (“POSA”)

would have a B.S. degree in chemistry or a related field and 2-5 years of experience in drug development that could include experience with peptide chemistry and/or peptide engineering. The POSA could also include individuals having a master’s degree or Ph.D. in chemistry or a related field with comparatively less experience in drug development involving peptide chemistry and/or peptide engineering. The POSA could have worked in consultation with individuals with knowledge and experience with the target drug receptor and of the disease condition to be treated. In particular, this team could include a clinical pharmacologist with experience with the target drug receptor (here, GCC receptors) or a medical doctor with experience in treating GI disorders, who may also have experience designing and running clinical trials, or a pharmaceutical formulator.

Resp. 25–26 (citing Ex. 2024 ¶¶ 30–32). However, Patent Owner’s expert states his analysis and conclusion would be the same under Petitioner’s definition of a POSA. Ex. 2024 ¶ 32.

The parties’ descriptions of a POSA are substantially similar. Indeed, neither side identifies any meaningful distinction between the two. Because we find Petitioner’s description of the level of ordinary skill in the art to be sufficiently supported by the record, we apply it for our analysis. That said, if we applied Patent Owner’s description, all of our findings and conclusions would be the same.

C. Claim Construction

Petitioner asserts that “[t]he claim terms do not require construction to apply the grounds.” Pet. 14. Patent Owner does not respond to this contention, or otherwise address claim construction in its Response. Accordingly, neither party has identified any claim term for construction.

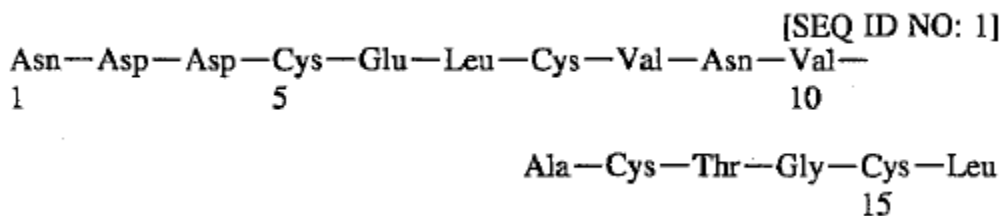
We agree that no formal claim construction is necessary for purposes of our Decision. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (explaining that it is only necessary to “construe terms ‘that are in controversy, and only to the extent necessary to resolve the controversy’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))).

D. Overview of the Cited References

i. Currie

Currie is a United States Patent that was issued on February 6, 1996. Ex. 1005, code (45). Petitioner contends, and Patent Owner does not dispute, that Currie qualifies as prior art under 35 U.S.C. § 102(b). Pet. 22.

Currie describes human uroguanylin. Ex. 1005, 1:47–63. Currie discloses the amino acid sequence for human uroguanylin in SEQ ID NO:1, which is reproduced below.



Id. at 1:50–55. The peptide shown in SEQ ID NO:1 of Currie is 16 amino acids long and has an Asp³.

Currie teaches that human uroguanylin “is an endogenous stimulator of intestinal guanylate cyclase” and “has been found to stimulate increases in cyclic GMP levels in a manner similar to guanylin and the STs.”¹¹ Ex. 1005, 2:6–8. Currie describes tests in which human uroguanylin “caused a concentration-dependent increase in T84 cell cyclic GMP” and “appeared to be more potent than human guanylin, but less potent than ST.” *Id.* at 6:11–14, Fig. 3A.¹² Currie also describes the results of a “competitive binding assay” showing “ST and human uroguanylin had similar affinities for the receptors on T84 cells and human guanylin had a much lower affinity.” *Id.* at 6:16–19, Fig. 3B. According to Currie these and other results described therein, suggest “human uroguanylin may . . . act as a laxative and be useful in patients suffering from constipation, e.g., cystic fibrosis patients who suffer severe intestinal complications from constipation.” *Id.* at 2:20–24.

ii. *Li*

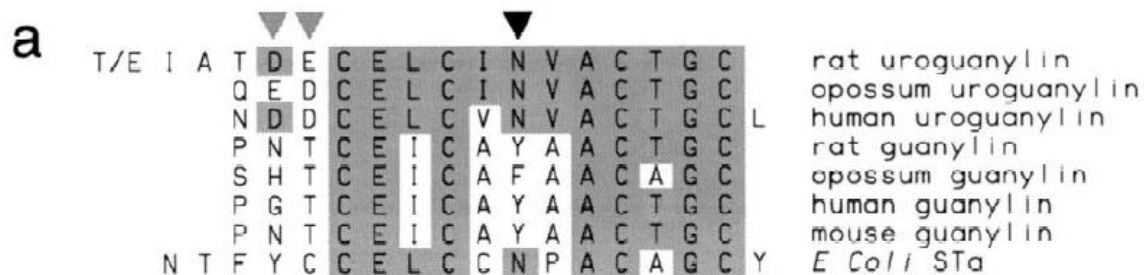
Li is an article from a scientific journal bearing a 1997 publication date. Ex. 1006, 45. Petitioner contends, and Patent Owner does not dispute, that Li qualifies as prior art under 35 U.S.C. § 102(b). Pet. 24.

Li describes the amino acid sequence for rat uroguanylin, explaining it has “high homology with uroguanylin” from humans and opossum uroguanylin. Ex. 1006, Abstr. Figure 6a of Li, reproduced below, is a

¹¹ Currie explains that “STs” are a “family of heat stable enterotoxins” produced by *E. coli* and other bacteria “that activate intestinal guanylate cyclase” and “cause secretory diarrhea.” Ex. 1005, 1:31–39.

¹² Although it refers to “Fig. 4a” and “Fig. 4b,” the results Currie describes appear to be those shown in Figures 3A and 3B. *Compare* Ex. 1005, 6:11–22, *with id.* at Figs. 3A–B.

sequence alignment comparing the amino acid sequences of uroguanylin, guanylin and ST from different species.



Id. at 52. Figure 6a shows the sequence for rat uroguanylin (first row) aligned with the sequences for opossum uroguanylin (second row) and human uroguanylin (third row), with shading added to indicate “amino acid identities” (i.e., positions where the sequences have the same amino acid).

Id. The alignment in Figure 6a shows that human and opossum uroguanylin have an aspartic acid (D) residue at position 3 in those sequences, whereas rat uroguanylin has a glutamic acid (E) at the corresponding position. Figure 6a also depicts “stippled arrowheads,” i.e., gray triangles, above the residues at positions 2 and 3 and a “solid arrowhead,” i.e., a black triangle, above the residue at position 9. *See id.* at 54 (left column). Li explains that these “arrowheads denote structural features described in the text.”

Li teaches that

[t]he affinity of GCC [i.e., guanylate cyclase type C receptor] for uroguanylin (opossum or human) is about 10-fold higher than its affinity for guanylin (rat or human). Thus, features that are found in uroguanylin, but not in guanylin, offer information about structural elements that specify the strength of the ligand/receptor interaction. Of particular interest are two residues that are basic or uncharged in guanylin but acidic in uroguanylin (stippled arrowheads), and one residue that contains an aromatic ring in guanylin but an acidic amide in uroguanylin (solid arrowhead).

Ex. 1006, 54 (left column). Li explains that rat uroguanylin “follows the consensus sequence of uroguanylin rather than that of guanylin” “[a]t all three positions” denoted by the arrowheads in Figure 6a “and thus we would expect its affinity to be comparable to that of opossum or human uroguanylin.” *Id.*

iii. *Narayani*

Narayani is an article from a scientific journal bearing a 1995 publication date. Ex. 1007, 39. Petitioner contends, and Patent Owner does not dispute, that Narayani qualifies as prior art under 35 U.S.C. § 102(b). Pet. 26.

Narayani describes tests of polymer-coated gelatin capsules. Ex. 1007, Abstr. Narayani explains that

[g]elatin capsules were coated with various concentrations of sodium alginate and cross-linked with appropriate concentrations of calcium chloride and tested *in vitro* for resistance to gastric and intestinal medium. Gelatin capsules coated with 20% w/v of the polymer which gave the most promising result *in vitro* were evaluated in human volunteers for their *in vivo* gastro intestinal tract behavior.

Id. Narayani teaches that “the alginate coated gelatin capsules remained intact as long as they were retained in the stomach (up to 3 h) and then migrated to the ileocecal region of the intestine and disintegrated.” *Id.* According to Narayani,

[t]he results of this study clearly suggested that alginate-coated gelatin capsules are safe candidates as oral delivery devices to carry microspheres containing bioactive peptides and proteins and dump them selectively in the large intestine where therapeutic action or drug absorption is desired. The

polymer-coated gelatin capsules will facilitate the routine use of the oral route of drug delivery for protein and peptide drugs.

Id. at 47.

iv. *Campieri*

Campieri is an article from a scientific journal bearing a 1997 publication date. Ex. 1008, 209. Petitioner contends, and Patent Owner does not dispute, that Campieri qualifies as prior art under 35 U.S.C. § 102(b). Pet. 27.

Campieri describes a clinical study involving the oral administration of certain steroids, i.e., prednisolone and budesonide, to treat Crohn's disease. Ex. 1008, Abstr. Campieri teaches that budesonide in "controlled ileal release (CIR) capsules . . . administered at 9 mg once daily or 4.5 mg twice daily, is comparable to prednisolone in inducing remission in active Crohn's disease." *Id.*

v. *Ekwuribe*

Ekwuribe is a United States Patent that issued on October 25, 1994. Ex. 1009, code (45). Petitioner contends, and Patent Owner does not dispute, that Ekwuribe qualifies as prior art under 35 U.S.C. § 102(b). Pet. 28.

Ekwuribe discloses "[a] stabilized conjugated peptide complex comprising a peptide conjugatively coupled to a polymer including lipophilic and hydrophilic moieties." Ex. 1009, Abstr. Ekwuribe discloses examples that incorporate polyethylene glycol (PEG) into the polymer. See *id.* at 13:4–52. Ekwuribe teaches that "[t]he presence of these PEG residues will impart hydrophilic properties to the polymer and to the corresponding polymer-peptide conjugates." *Id.* at 12:41–43; 14:50–55.

E. Ground 1: Obviousness over Currie and Li

Petitioner contends that claim 1 would have been obvious over the combination of Currie and Li. *See* Pet. 32–39. More specifically, Petitioner contends that Currie discloses the amino acid sequence for human uroguanylin, which differs from the claimed SEQ ID. NO:20 in only respect. That is, in the human uroguanylin sequence there is an aspartic acid residue at position 3 (“Asp³”), whereas SEQ ID NO:20 has a glutamic acid residue at positions 3 (“Glu³”). Petitioner contends that it would have been obvious to modify uroguanylin by substituting glutamic acid for the aspartic acid at position 3 (i.e., “the Glu³ substitution”), thereby resulting in the peptide of claim 1.

Patent Owner does not dispute that the only difference between the uroguanylin sequence disclosed in Currie and the peptide of claim 1 is that uroguanylin has an Asp³, whereas the recited peptide has a Glu³. Rather, Patent Owner argues that it would not have been obvious to modify Currie’s uroguanylin sequence as Petitioner contends. *See* Resp. 26–66.

The parties’ disputes are focused on three issues: (1) whether it would have been obvious to select uroguanylin as a lead compound for modification; (2) assuming uroguanylin was selected as a lead, whether it would have been obvious to make the Glu³ substitution; and (3) Patent Owner’s unexpected results showing, which it offers as objective evidence of the alleged non-obviousness of the recited peptide. We address each of these issues below before reaching an ultimate conclusion on obviousness.

i. *Whether a POSA would have had a reason to select human uroguanylin as a lead compound*

Petitioner contends that a POSA would have recognized uroguanylin “as a good candidate for oral administration for delivery to the intestines to treat constipation” and therefore selected it as a lead compound for further modification. See Pet. 19 (citing Ex. 1002 ¶¶ 59–61, 82–86, 89–93, 107).

Patent Owner responds to Petitioner’s contentions, arguing that “a POSA would not have selected human uroguanylin as a lead compound over heat-stable enterotoxins.” Resp. 29. Patent Owner explains that “[h]eat stable enterotoxins (STs) are secretory peptides produced by some bacterial strains, such as enterotoxigenic *E. coli*.” *Id.* at 13. According to Patent Owner, STs were known to be more stable than other GCC agonist peptides and “exhibited a potency and binding affinity that surpassed uroguanylin and guanylin in the acidic and alkaline pHs of both the small and large intestines.” *Id.* at 14.

In addition, Patent Owner asserts that “human uroguanylin was plagued by isomeric instability,” i.e., it “was known to have two topological interconverting isomers: Isomer A, which was known to be biologically active, and Isomer B, which was known to be biologically inactive.” Resp. 29 (citing Ex. 2011, 27; Ex. 2020, 223). According to Patent Owner,

[b]ecause of human uroguanylin’s known interconversion/topoisomerism dilemma and its attendant problems with manufacture, formulation, storage and *in vivo* unpredictability, a POSA would not have selected human uroguanylin as a lead compound. Instead . . . the heat-stable enterotoxins would have been a far more promising option, for they were not afflicted with topoisomerism, were more biologically effective, and were pH-independent, giving them greater versatility throughout the small intestine and colon.

Id. at 32–33.

Petitioner disputes Patent Owner’s arguments against the selection of human uroguanylin as a lead compound for developing synthetic analogues. *See* Reply 4–11. According to Petitioner, “[u]roguanylin’s known properties were very promising for synthesizing an analogue” and “Currie acknowledged enterotoxins’ higher potency yet identified uroguanylin as a very promising lead compound for further modification.” *Id.* at 4–5 (citing Ex. 1063 ¶¶ 10–12). Petitioner contends that a POSA would have understood that “maximizing potency beyond uroguanylin was not necessarily desirable” because “replicating enterotoxin’s potency and lack of pH sensitivity might cause severe diarrhea and even death.” *Id.* at 6–7 (citing Ex. 1063 ¶¶ 16–18; Ex. 1064 ¶¶ 42–46; Ex. 1062, 62:5–64:13, 76:13–79:20; Ex. 1071).

Petitioner also asserts that topoisomerism would not have led a POSA away from selecting uroguanylin as a lead compound. *See* Reply 7–11. According to Petitioner, uroguanylin was known to freely convert between topoisomers at acidic pH of 4.5, but a POSA “would have expected oral uroguanylin not to be exposed to this low pH for over an hour, with no more than 1% interconversion” and therefore “[u]roguanylin was known as stable and effective *in vivo* after oral administration.” *Id.* at 7 (citing Ex. 1063 ¶ 27; Ex. 1018, G641–G642; Ex. 2021, 2:28–3:1). Moreover, Petitioner asserts that Patent Owner’s arguments about potential purification and storage problems are overstated because “purifying topoisomers was straightforward” and “administering pharmaceuticals as mixtures of interconverting active and inactive forms was common, e.g., racemic drugs like ibuprofen.” *Id.* at 8–10 (citing Ex. 1063 ¶¶ 34, 40, 42).

In its Sur-reply, Patent Owner disputes Petitioner's evidence that uroguanylin would only be expected to spend an hour exposed to pH 4.5 *in vivo* because it is based "on gastrointestinal pHs and transit times in *healthy* patients, not constipated patients." Sur-reply 3. According to Patent Owner, "Petitioner also downplays the difficulties with purifying, formulating, and storing human uroguanylin—difficulties a POSA would have considered when selecting a lead compound." *Id.* at 4–5. Thus, Patent Owner contends that uroguanylin's "topoisomeric instability" and "inferior potency" presented unnecessary challenges, which a POSA would have avoided by instead selecting a heat-stable enterotoxin as a lead compound. *See id.* 2–3.

We find that Petitioner has the better argument on this issue. As Petitioner points out, it "need only identify 'some reason' that would have sufficed for a chemist to modify" human uroguanylin. Reply 3 (quoting *Bristol-Myers Squibb Co. v. Teva Pharms. USA, Inc.*, 752 F.3d 967, 973 (Fed. Cir. 2014)); *see also Otsuka Pharm. Co. v. Sandoz, Inc.*, 678 F.3d 1280, 1292 (Fed. Cir. 2012) ("In determining whether a chemist would have selected a prior art compound as a lead, the analysis is guided by evidence of the compound's pertinent properties.").

Here, Currie itself provides that reasoning. Currie specifically identifies uroguanylin as a potential "laxative" that may "be useful in patients suffering from constipation, e.g. cystic fibrosis patients who suffer with severe intestinal complications from constipation." Ex. 1005, 2:20–24. Currie also teaches that STs affect the same biological pathway, but are derived from pathogenic bacteria and are dangerous. *Id.* at 1:30–44. Thus, Currie's teachings and data showing that uroguanylin exhibits activity that is intermediate between the higher activity associated with STs and the lower

activity associated with guanylin (*see, e.g., id.* at Figs. 3A–B) support Dr. Peterson’s testimony that a POSA would have recognized uroguanylin as a candidate for “treating constipation using a more controlled version” of the same pathway targeted by STs. Ex. 1002 ¶¶ 59–60. This evidence is sufficient to show that a POSA would have been motivated to select human uroguanylin as a lead compound.

Patent Owner’s evidence that topoimerism was a known concern with human uroguanylin does not negate Petitioner’s rationale for selecting uroguanylin as a lead compound. *See* Resp. 29–32 (citing Ex. 2010, 236; Ex. 2011, 27, 30; Ex. 2020, 223; Ex. 2024 ¶¶ 49–60, 127–34). Rather, it suggests that isomeric instability was a challenge, but not one that would discourage further investigation of uroguanylin analogues. In this regard, we credit the testimony of Petitioner’s declarants that a POSA would have understood that the negative effects caused by *in vivo* interconversion could be mitigated by using known techniques, e.g., formulating a tablet to target release in the higher pH environment of the intestines as opposed to the lower pH environment of the stomach. *See* Ex. 1063 ¶¶ 24–29; Ex. 1064 ¶¶ 24–32 (explaining, *inter alia*, that the “GCC receptor that uroguanylin activates is concentrated in the proximal small intestines,” an “enteric coating” can be used to “protect[] the tablet from the stomach fluids,” and that “if exposure to prolonged acidic conditions were a concern there were routine approaches” that could be taken to address such).

Finally, Patent Owner’s argument that STs would have been viewed as a better choice for a lead compound does not overcome Petitioner’s showing that human uroguanylin would also have been a “natural choice for further development efforts.” *See Bristol-Myers*, 752 F.3d at 973 (quoting

Altana Pharma AG v. Teva Pharms. USA, Inc., 566 F.3d 999, 1008 (Fed. Cir. 2009)). The Federal Circuit has cautioned that it would be overly restrictive to view the “lead compound test” to require “that the prior art must point to only a single lead compound for further development.” *Altana*, 566 F.3d at 1008. Here, Petitioner articulates reasoning supported by evidence to explain why a POSA would have selected uroguanylin for further development in view of both its advantages and disadvantages relative to STs. *See* Ex. 1002 ¶¶ 59–61, 89–90; Ex. 1005, 1:30–44, 2:20–24; Ex. 1063 ¶¶ 16–21. We find Petitioner’s showing sufficient to meet its burden even if a POSA might also have viewed STs as another class of promising lead compounds.

ii. *Whether it would have been obvious to make the Glu³ substitution*

Petitioner argues that it would have been obvious to make the Glu³ substitution to human uroguanylin. More specifically, Petitioner, relying on testimony from Dr. Peterson, asserts that: (1) aspartic acid and glutamic acid are structurally similar such that substituting glutamic acid for aspartic acid would have been regarded as a “conservative substitution,” which was a “primary strategy employed by protein engineers” (Ex. 1002 ¶¶ 73–75); (2) Li teaches that the presence of an acidic residue at position 3 in human uroguanylin is of “particular interest” to the ligand/receptor interaction and its sequence alignment shows that rat uroguanylin has a glutamic acid at the corresponding position, both of which would suggest substituting glutamic acid for the Asp³ in human uroguanylin (*id.* ¶¶ 137–152); (3) a POSA would have expected that the peptide resulting from this substitution “would be comparably more protonated than native human uroguanylin . . . in the large

intestine and intestinal mucosa” and therefore it would have been reasonably expected to have “improved activity” as compared to the native human uroguanylin (*id.* ¶¶ 165–170); and (4) a POSA would have understood that another benefit of this substitution would be that it eliminates known sources of aspartimide formation during synthesis (*id.* ¶¶ 175–179). *See* Pet. 34–39.

Patent Owner argues that “a POSA would not have substituted Asp³ with Glu³ with a reasonable expectation of success” because nothing in the art suggested that doing so would have addressed uroguanylin’s interconversion issue or improved its activity. Resp. 38. Patent Owner asserts that “[i]f anything, a POSA would have inserted a third disulfide bridge,” i.e., made a different modification to the peptide,¹³ because “it was known from the heat-stable enterotoxins that the third disulfide bridge made them more conformationally rigid and stable (preventing their interconversion) and also beneficially led to a ‘more efficient interaction with the receptor’ and ultimately their higher potencies.” *Id.* at 39 (quoting Ex. 2010, 230, 235; Ex. 2024 ¶¶ 145–146).

Patent Owner also disputes Petitioner’s argument that a POSA would have been motivated to make the Glu³ substitution because it is a “conservative substitution.” *See* Resp. 41–51. According to Patent Owner, Petitioner’s conservative substitution rationale “runs contrary to the lead

¹³ Disulfide bridges are formed between cysteine residues. For example, SEQ ID NO: 20 in the Specification shows a first disulfide linkage (denoted by an “*”) between the cysteines at positions 4 and 12 and a second disulfide linkage between the cysteines at positions 7 and 15 (denoted by an “**”). Ex. 1001, 5:6–16; Ex. 1002 ¶ 23. Therefore, adding a third disulfide bridge would involve modifications to add cysteine residues rather than a Glu³ substitution.

compound analysis,” which requires an expectation that the lead compound could be modified “to improve upon its activity.” *Id.* at 41–42 (internal quotations omitted). Instead, urges Patent Owner, “[i]t was understood that a conservation substitution ‘of an amino acid with another amino acid with similar physicochemical properties will not influence the stability and function of the protein.’” *Id.* at 43 (quoting Ex. 2035, 397). Patent Owner points to Jonson (Ex. 2035), which was “published shortly before 2002 [and] stated that ‘very different spatial preferences were found for glutamic acid and aspartic acid’” and “[t]he common belief that a glutamic acid to aspartic acid mutation is conservative is contrary to the observations shown.” *Id.* (quoting Ex. 2035, 397, 400). Patent Owner also cites references it contends show the importance of Asp³ to human uroguanylin’s activity to suggest that a POSA would not have been motivated to make a change to that residue. *See id.* at 49–51 (citing Ex. 1021; Ex. 2020; Ex. 2021; Ex. 2041).

Patent Owner further contends “Petitioner reads more into Li than the POSA would have objectively understood Li to convey” because Li is directed primarily to rat uroguanylin. Resp. 46 (citing Ex. 2024 ¶¶ 160–165). According to Patent Owner, “[a] POSA would not have looked to rat uroguanylin to improve upon human uroguanylin” because “Li’s experimental data demonstrated that rat uroguanylin (fraction 16 in Fig. 3) was less active than all other GCC agonists tested” and rat uroguanylin was “known to suffer from topoisomerism to an even greater extent than human uroguanylin.” *Id.* at 47 (citing Ex. 1006, 49 (Fig. 3); Ex. 2011, 30). Moreover, Patent Owner argues that “the affinities of uroguanylin and guanylin were known to depend strongly on pH, with uroguanylin showing

increased affinity at acidic pH and guanylin demonstrating increased affinity at alkaline pH.” *Id.* at 48 (citing Ex. 1021, 2075). According to Patent Owner, a POSA with the “motivational objective” of treating constipation “would not have read Li and looked to modify human uroguanylin with rat uroguanylin but, if anything, would have looked to either the heat-stable enterotoxins or guanylin.” *Id.* at 49 (citing Ex. 2024 ¶¶ 165, 188–192).

Turning to the Petitioner’s other arguments for making the Glu³ substitution, Patent Owner asserts that the argument that a Glu³ substitution would have been expected to result in a peptide with better activity because glutamic acid has a higher pKa than aspartic acid is “fundamentally flawed.” Resp. 52. According to Patent Owner, “Petitioner and Dr. Peterson rely exclusively on the pKa values for the side-chain groups of Asp and Glu as **free amino acids**” even though “Dr. Peterson admitted, ‘it’s not accurate to compare a free amino acid pKa necessarily with a peptide amino acid pKa.’” *Id.* (quoting Ex. 2026, 101:15–102:2). Patent Owner argues “[t]he prior art establishes that Asp could actually deprotonate at a higher pH” than Glu depending on its interactions with “the surrounding amino acids in the peptide chain” and “environmental milieu,” undermining this aspect of Petitioner’s rationale for making the Glu³ substitution. *Id.* at 53–54 (citing Ex. 2024 ¶¶ 182–186; Ex. 2045, 37 (Table 1)).

Moreover, Patent Owner argues that the additional benefit of avoiding aspartimide formation during synthesis would not have motivated a POSA to make the Glu³ substitution because “such an assertion . . . would result in the categorical nonuse of Asp in peptide development—an absurd result.” Resp. 56. Patent Owner urges that “Dr. Peterson testified that the prior art reported aspartimide formation ‘for glycine and asparagine,’ which are

amino acids in human uroguanylin that Dr. Peterson does not believe should be replaced.” *Id.* (quoting Ex. 2026, 119:14–121:3). Therefore, according to Patent Owner, a “POSA would not have been motivated to substitute an amino acid residue to avoid a source of aspartimide, which could be easily avoided with protecting groups or removed through purification.” *Id.* at 57.

Petitioner replies to Patent Owner’s argument that a POSA would have been motivated to add a third disulfide bridge, noting that “even if another modification was also obvious, a POSA had a good reason to make [Glu3]-uroguanylin.” Reply 13 (citing Ex. 1063 ¶¶ 50, 55). Moreover, Petitioner asserts that “Marx observed . . . the ionizable side chains of Asp², Asp³, and Glu⁵” in uroguanylin “‘may be involved in the control of stabilization of the two [topo]isomers.’” *Id.* at 13 (quoting Ex. 2010, 236, 238). Thus, according to Petitioner, “[a] POSA considering topoisomeric conversion . . . had additional reason for the [Glu³]-substitution, an ionizable side-chain modification at position 3.” *Id.* at 14.

Petitioner replies to Patent Owner’s citation to Jonson, urging that it “‘found an exception in a *different* peptide to the ‘common belief that a Glu to Asp mutation is conservative’ because side-chain length mattered for *tryptophan*” which is not present in uroguanylin. Reply. 15 (citing Ex. 2035, 400). According to Petitioner, “Jonson confirms replacing Asp with Glu generally retains a peptide’s activity, with the correlation being ‘particularly’ true for ‘charged residues’ like Asp and Glu.” *Id.* at 15–16 (quoting Ex. 2035, 401, Fig. 4).

Regarding Patent Owner’s other citations purporting to show the importance of Asp³ to uroguanylin’s activity, Petitioner explains that “***deleting*** Asp³ is not the same as ***substituting*** Glu³.” *See id.* at 16–17 (citing

Ex. 1063 ¶¶ 90–91). According to Petitioner, Li and other prior art taught “only that all uroguanylin peptides have aspartate *or* glutamate residues at these positions and these acidic residues should not be **deleted**.” *Id.* at 17 (internal quotations omitted); *see also id.* at 19 (explaining that while Ex. 2021 identifies Asp² and Asp³ as “highly conserved,” the authors did not mark those residues in italics as they did with other residues they “believe[d] are essential for activity”).

Petitioner also contends that Patent Owner “is wrong to say Li teaches rat uroguanylin was less potent because of the Glu³-substitution.” Reply 17. According to Petitioner, “[t]he experiments in Li Fig. 3” that Patent Owner relies upon “are not designed to, and do not, determine that rat uroguanylin is less potent” because the rat uroguanylin in fraction 16 was “extracted from rat intestines” and compared to synthetic standards for other peptides “merely to illustrate that preincubation provided no activity benefit—similar to fraction 16.” *Id.* at 18 (citing Ex. 1063 ¶¶ 79–87). Thus, urges Petitioner, “Li does not even use identical amounts in each assay and provided no opportunity to compare the different peptides’ potency.” *Id.* (citing Ex. 1063 ¶¶ 86–87).

Petitioner replies to Patent Owner’s evidence that the R group pKa of free amino acids is different from their pKa in the peptide, urging that evidence “relate[s] to residues buried inside a peptide” that “are not analogous to the Asp³ or Glu³ pKas in plecanatide,” which “were known to be not buried.” Reply 20 (citing Ex. 1063 ¶¶ 101–103; Ex. 2043, 2; Ex. 2044, 567–570 (Table 1); Ex. 2045, 34–35, 37 (Table 1)). According to Petitioner, “Dr. Peterson never argued pKa **values** would be identical in the peptide chain,” but rather that “the relative **difference** between the pKas of

the two amino acids in the peptide at the solvent-exposed, unburied position 3 of uroguanylin would be very similar to the difference in the free amino acids” and that this difference would have motivated a POSA to make a Glu³ substitution. *See id.* (citing Ex. 1063 ¶¶ 98–100, 104; Ex. 2026, 100:3–13).

Regarding the motivation to avoid aspartimide formation, Petitioner replies that “the problem is **pairing** (e.g., of aspartate or cysteine) on an aspartate’s amino end, not these residues’ mere existence.” Reply 21 (citing Ex. 1063 ¶ 111). According to Petitioner, Patent Owner’s argument that “aspartimide impurities might be removed through purification or reduced using additional protecting groups . . . cannot teach away from an otherwise-favorable substitution” that would have avoided this undesirable pairing of residues. *See id.* at 21–22 (citing Ex. 1063 ¶¶ 112–113; Ex. 1024, Abstr., 202 (Table 2); Ex. 1022, 64; Ex. 1023, 107, 111).

In its Sur-reply, Patent Owner generally disputes Petitioner’s contentions above, continuing to urge that “a POSA would not have been motivated to make the ‘specific molecular modifications’ to human uroguanylin with a reasonable expectation of achieving plecanatide.” *See* Sur-reply 9–18 (quoting *Takeda Chem. Indus., Ltd. v. Alphapharm Pty., Ltd.*, 492 F.3d 1350, 1356–57 (Fed. Cir. 2007)).

Upon considering the parties’ arguments and evidence, we find that Petitioner’s conservative substitution rationale demonstrates a motivation to make the Glu³ substitution to human uroguanylin with a reasonable expectation of success. We are not, however, persuaded by the other rationales, i.e., Petitioner’s pKa, aspartimide formation, and topoisomerism arguments, Petitioner contends would have motivated that modification.

Petitioner posits that it would have been obvious to make the Glu³ substitution because it was a “conservative substitution,” i.e., a single substitution of one acidic residue (Asp) for the only other residue with a carboxylic acid in its side chain (Glu) based on Li’s teaching that the presence of certain acidic residues in uroguanylin, as opposed to the basic or uncharged residues found at the corresponding positions in guanylin, is “[o]f particular interest.” Pet. 34 (quoting Ex. 1006, 54) (emphasis omitted). Li supports that rationale. More specifically, it teaches that uroguanylin has a “10-fold higher” affinity for GCC than guanylin and therefore the “features that are found in uroguanylin, but not in guanylin, offer information about structural elements that specify the strength of the ligand/receptor interaction.” Ex. 1006, 54. Li identifies those elements by marking them with arrowheads in Figure 6a below. *Id.* at 52.

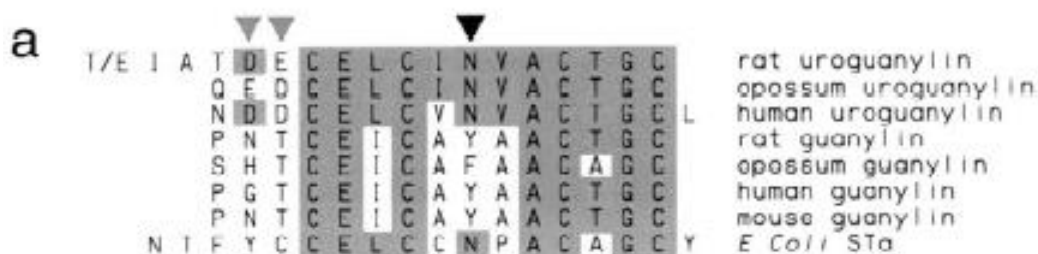


Figure 6a is a sequence alignment of different guanylin, uroguanylin and ST peptides with the amino acid sequence for human uroguanylin in the third row from the top. *Id.* Figure 6a includes “stippled arrowheads” marking two positions in the alignment, which correspond to the Asp (D) residues at positions 2 and 3 in human uroguanylin.¹⁴ *Id.* at 54. The rat and opossum uroguanylin sequences have either an Asp or Glu (E) at the corresponding

¹⁴ Figure 6a also includes an arrowhead over the residue (N) at position 9 in human uroguanylin, thereby identifying it as another position of interest.

positions. *Id.* at 52. Li teaches that the fact that these residues are “acidic” (i.e., either Asp or Glu) in all of the uroguanylin sequences is one of the features that specifies uroguanylin’s higher affinity for GCC. *Id.* at 54. Thus, Li both highlights position 3 in human uroguanylin as one of three positions of interest and suggests that if a change were made there it should be the substitution of another acidic residue, i.e., the Glu³ substitution. In addition, Li evidences that a POSA would have reasonably expected that the Glu³ substitution could be successfully tolerated because rat uroguanylin already has a Glu at the corresponding position and is “expect[ed]” to have affinity “comparable to that of opossum or human uroguanylin.” *Id.*

We disagree with Patent Owner’s argument that the data in Figure 3 of Li would have suggested that rat uroguanylin was less potent than uroguanylins from other species. According to Patent Owner, fraction 16 in Li’s Figure 3 is rat uroguanylin. Resp. 47 (citing Ex. 1006, 49, Fig. 3). Fraction 16, as well as the other peptide-containing fractions in Figure 3, were isolated from rat intestines, whereas the data for guanylin, uroguanylin, and STa in that figure comes from testing of synthetic standards. See Ex. 1006, 46 (describing preparation of fractions from rat intestines), 49 (Fig. 3 caption describing data for “synthetic” guanylin and uroguanylin), 50–51; Ex. 1060, 167:12–168:8; Ex. 1063 ¶¶ 84–85. Dr. Peterson persuasively testifies that a POSA would not have compared data from an isolated peptide with a synthetic standard to assess their relative activity. Ex. 1063 ¶¶ 81–82, 85–86; see also Ex. 1060, 144:8–20 (Dr. Davies acknowledging on cross-examination that one would not use a synthetic standard in such an assay). We also credit Dr. Peterson’s testimony that there is no way to know whether fraction 16 contained a comparable amount of peptide to the

synthetic standards tested in Figure 3 and therefore the data there do not show that fraction 16 was more or less potent than opossum or human uroguanylin. Ex. 1063 ¶¶ 87, 89. Indeed, Patent Owner’s argument that the data in Figure 3 show that rat uroguanylin is less potent is expressly contradicted by Li’s other teachings. That is, Li’s authors stated that they “expect [rat uroguanylin’s] affinity to be comparable to that of opossum or human uroguanylin” and that “[d]ose/response curves with synthetic rat peptide will be required to test this idea directly.” Ex. 1006, 54. Thus, we find Dr. Peterson’s testimony above more credible than the competing testimony of Dr. Davies (see Ex. 2024 ¶¶ 79, 164) on this point.

While Li does not suggest that rat uroguanylin is less potent than uroguanylins from other species, there is also nothing in the record that suggests a POSA would have expected rat uroguanylin to have greater potency. See Tr. 15:7–16:7 (Petitioner’s counsel acknowledging there is no such evidence in the record); Ex. 1063 ¶ 82 (explaining that Li’s tests “do not permit comparison of the affinity or potency of rat uroguanylin to human or opossum uroguanylin”). Thus, Petitioner has not shown that a POSA would have expected the Glu³ substitution to improve potency or any other property of the resulting peptide. Instead, Li merely evidences a reasonable expectation that the peptide resulting from the Glu³ substitution would retain activity similar to human uroguanylin. This supports Dr. Peterson’s testimony, which we credit, that a POSA would understand this to be a “conservative substitution.” See Ex. 1002 ¶ 76 (explaining that when orthologs from “related species have different amino acids at a particular position (variant residues), this often indicates that the particular observed differences preserve the functionality of the proteins” and they “often will

differ by *conservative substitutions*, that is, substitutions involving similar amino acid residues”) (internal quotations omitted).

Contrary to Patent Owner’s assertion, Petitioner’s “conservative substitution” rationale for making the Glu³ substitution is consistent with precedent. *See* Resp. 41–42. The Federal Circuit has explained that the rationale for generating analogs of a lead compound may simply be an expectation that the resulting compound will exhibit “similar properties.” *See Bristol-Myers*, 752 F.3d at 973 (quoting *Otsuka*, 678 F.3d at 1293); *In re Dillon*, 919 F.2d 688, 692 (Fed. Cir. 1990) (en banc)). For example, in *Bristol-Myers* the expert testimony showed that “[a]fter selecting a lead compound” a POSA “would seek to make small, conservative changes to that structure” because doing so was common in drug development. 752 F.3d at 974. Here, Dr. Peterson offers analogous testimony, which we credit, explaining that making conservative substitutions was a “primary strategy” that protein engineers at the time used to develop synthetic analogs of naturally-occurring peptides. Ex. 1002 ¶¶ 73–75. Moreover, Dr. Peterson persuasively testifies that it was a routine matter for a POSA to synthesize and characterize the properties of such synthetic analogs. Ex. 1002 ¶¶ 66–67, 106, 130–131; Ex. 1063 ¶ 14; *see also* Ex. 1005, 3:8–45 (teaching that human uroguanylin can be prepared by conventional peptide synthesis methods).

Patent Owner’s remaining arguments against Petitioner’s conservative substitution rationale are also unavailing. Patent Owner cites Jonson’s statement that “very different spatial preferences were found for glutamic acid and aspartic acid” and “[t]he common belief that a glutamic acid to aspartic acid mutation is conservative is contrary to the observations

shown.” Resp. 43 (quoting Ex. 2035, 397, 400). But those observations relate to “tryptophan-glutamic acid” and “tryptophan-aspartic acid” pairs, which a POSA would have understood to be inapplicable to human uroguanylin because it does not contain any tryptophan residues. Ex. 2035, 400. Indeed, contrary to Patent Owner’s position, Jonson provides further evidence that conservative substitution was a common strategy for making peptide analogs. *See id.* at 397 (“In protein engineering the concept of conservative mutations is frequently used. The general idea is that a substitution of an amino acid with another amino acid with similar physico-chemical properties will not influence the stability and function of the protein.”).

Patent Owner’s arguments about other “substituting options” are likewise unavailing. *See* Resp. 38–39 (urging that a POSA would have been motivated to add a third disulfide bridge), 44–45 (asserting there are “approximately 20¹⁶” substitutions that could be made to the 16 amino acids in human uroguanylin). It is true that if a POSA were to consider using any one of the 20 standard amino acids at any of the 16 positions in human uroguanylin there would be a huge number of possible variations. However, as explained above, Li highlights a handful of interesting positions with arrowheads, including human uroguanylin’s Asp³, and suggests that the acidic side chain at that position “specifies” uroguanylin’s higher affinity as compared to guanylin. Ex. 1006, 52 (Fig. 6A), 54. Thus, we credit Dr. Peterson’s testimony that Li substantially narrows the pool of substitutions a POSA could have made in the abstract to a finite set of conservative substitutions, which includes the Glu³ substitution, that a POSA would have been motivated to make. *See* Ex. 1002 ¶¶ 148, 156, 161

(testifying that a POSA's "attention immediately would have been drawn" to the positions identified in Li and that "[a]spartic and glutamic acid are the only two amino acids that possess an acidic side chain"). The fact that there may be some other obvious changes to try, e.g., a Glu² substitution¹⁵ or a modification to add a third disulfide bridge as Patent Owner urges (*see* Resp. 39–40), does not undermine Petitioner's showing that a POSA would have been motivated to make the Glu³ substitution with a reasonable expectation of success. *See Bristol-Myers*, 752 F.3d at 975–76 (noting other possible modifications a POSA might make to the lead compound, but determining that "the steps of deciding which bond to modify and how to modify that bond equate to a small, finite number of changes to try") (internal quotations omitted).

We also disagree with Patent Owner's argument that a POSA would not have made the Glu³ substitution because certain references taught the importance of Asp³ to human uroguanylin's activity. *See* Resp. 49–51 (citing Ex. 1021 (Hamra 1997); Ex. 2020 (Klodt); Ex. 2021 (WO '266)). Hamra 1997 studied synthetic analogs of opossum uroguanylin wherein the three *N*-terminal residues (Gln-Glu-Asp) had been deleted, finding that deleting those residues reduced binding affinity under acidic conditions. *See* Ex. 1021, 2709. According to Patent Owner, Klodt "[s]imilarly" reported that "deleting Asp 3 in human uroguanylin left it with very little activity." Resp. 49 (citing Ex. 2020, 228 (Fig. 4)). But we agree with Petitioner that

¹⁵ As shown above, Li's Figure 6 marks both the Asp² and Asp³ in human uroguanylin with stippled arrowheads. For this reason, Petitioner acknowledges that Li also evidences a motivation to make the Glu² substitution. *See* Pet. 38–39.

“***deleting*** Asp³ is not the same as ***substituting*** Glu³.” Reply 16. Indeed, if anything, these references support Li’s teaching that the presence of an acidic residue, either Asp or Glu, at position 3 in human uroguanylin is one of the features that specifies uroguanylin’s higher affinity relative to guanylin. See Ex. 1006, 54. Thus, they do not undermine Petitioner’s conservative substitution rationale, which is premised on that teaching.

The statements Patent Owner points to in WO ’266 are not enough to overcome Petitioner’s showing. WO ’266 teaches that uroguanylin peptides have been identified from a variety of species, including humans and rats and “[t]he functionally active domain in most of these peptides is highly conserved.” Ex. 2021, 17:21–30. This teaching accords with Li’s observation that the uroguanylin sequences in its Figure 6A all share acidic residues at the positions corresponding to positions 2 and 3 in human uroguanylin. See Ex. 1006, 54. Moreover, while the “preferred embodiment” taught and claimed in WO ’266 retains Asp² and Asp³ while permitting modifications at certain other positions (see Ex. 2021, 13:20–33, 29:2–16), the preferences of the inventors in that patent application do not overcome Petitioner’s evidence showing that a POSA would have regarded the Glu³ substitution as a conservative substitution to try. Indeed, as Petitioner points out, Asp² and Asp³ are *not* among the residues marked with bold and italic letters in Figure 7, which WO ’266 teaches are the “residues . . . believed to be required for the functional activity of these peptides.” *Id.* at 7:26–8:3, Fig. 7. Thus, like Hamra 1997 and Klodt, WO ’266 does not undermine Petitioner’s showing that a POSA would have understood the Glu³ substitution to be a conservative substitution and therefore would have been motivated to make that modification to human uroguanylin.

While the record supports Petitioner's conservative substitution rationale, the other rationales Petitioner offers for making the Glu³ are not persuasive. First, Petitioner's argument that the Glu³ substitution "would have been expected to result in a protonated glutamate at a higher pH (pK_a = 4.25 rather than aspartic acid's 3.65) for better activity in the less acidic environment of the intestinal lumen" is based on the pK_a values for the R group of the free amino acid. See Pet. 36 (citing Ex. 1002 ¶¶ 157–61; Ex. 1012, 118, Table 5-1); Ex. 2024 ¶ 174. However, both sides' experts have since testified that it is not accurate to compare free amino acid pK_a with the pK_a when they are incorporated into the peptide chain because the latter will be affected by its surrounding environment. Ex. 2026, 101:12–102:2; Ex. 2024 ¶¶ 176–78. Indeed, the record reflects that "the pK_a value for an ionizable R group can change somewhat when an amino acid becomes a residue in a peptide" because "[t]he loss of charge in the α-carboxyl and α-amino groups, interactions with other peptide R groups, and other environmental factors can affect the pK_a." Ex. 1012, 127. For this reason, the same text Petitioner cites to show the pK_a values of aspartic and glutamic acid cautions that those values "cannot be strictly applied to peptides." *Id.*

Petitioner's argument that the "relative ***difference*** between the pK_as of the two amino acids in the peptide . . . would be very similar to the difference in the free amino acids" is unpersuasive. Reply 20 (citing Ex. 1063 ¶¶ 98–100, 104; Ex. 2026, 100:3–13). Patent Owner offers evidence showing that the R group pK_a of aspartic acid and glutamic acid in peptides can vary widely from the pK_a of the free amino acid. Ex. 2043, 2; Ex. 2044, 567–70 (Table 1). Patent Owner's evidence includes an instance

in which the pKa of aspartic acid's R group was the same or even higher than that of glutamic acid when incorporated into a peptide, which directly undermines Petitioner's position that the relative difference would be the same. Ex. 2045, 37 (Table 1). Petitioner dismisses this evidence, urging that it relates to "residues buried inside a peptide" that "are not analogous to the Asp³ or Glu³ pKas in plecanatide." Reply 20. But that misses the point. The record shows there are interactions beyond just whether the residue is buried that can affect the pKa. *See, e.g.*, Ex. 1012, 127 ("interactions with other peptide R groups, and other environmental factors can affect the pK_a"); Ex. 2069, 96:19–97:2, 97:19–98:1 (Dr. Peterson acknowledging that intrachain hydrogen bonds can affect pKa). Neither Petitioner, nor Dr. Peterson, sufficiently address these other interactions, much less do they do so within the context of the particular amino acid sequences at issue here. Accordingly, we credit Dr. Davies's testimony that a POSA "would have understood that the pKa values of amino side chain groups could not be predicted when the amino acid was incorporated into a peptide" and therefore would not have expected the Glu³ substitution to result in a peptide that would "remain protonated at a higher pH as compared to Asp or result in better activity in less acidic environments." Ex. 2024 ¶¶ 178–186.

Second, Petitioner's argument that a POSA would have been motivated to make the Glu³ substitution to avoid aspartimide formation is unpersuasive. According to Petitioner, changing the peptide to eliminate Asp pairs in its primary structure would have been an obvious solution to avoid the known problem associated with aspartimide formation during synthesis. However, each of the references Petitioner cites teach a different solution—the use of protecting groups—to avoid aspartimide without changing the

target molecule. Ex. 1022, 63; Ex. 1023, 111; Ex. 1024, 204; Ex. 2024 ¶¶ 198–201 (analyzing these references). Moreover, as Patent Owner points out, Dr. Peterson testified on cross-examination that the presence of other amino acids (e.g., glycine and asparagine) can also cause aspartimide formation. Resp. 56 (citing Ex. 2026, 119:14–19). Petitioner does not explain why a desire to avoid aspartimide formation would have motivated the Glu³ substitution, but not the substitution of other residues in human uroguanylin that might lead to the same problem. See Reply 21–22. For these reasons, we credit Dr. Davies’s testimony that a POSA “would not have been motivated to substitute an amino acid residue to avoid a source of aspartimide, which could be easily avoided with protecting groups.” Ex. 2024 ¶ 203.

Petitioner replies that “alternatives” like protecting groups “cannot teach away from an otherwise-favorable substitution.” Reply at 21. But again, this misses the point. It is Petitioner’s burden to show that its rationale would have motivated the Glu³ substitution. Patent Owner need not show that the prior art teaches away from such. At most, Petitioner shows that the Glu³ substitution might provide some *post hoc* benefit by simplifying synthesis of the modified peptide—not that a desire to avoid aspartimide formation would have motivated a POSA to change the peptide itself by making the Glu³ substitution.

Finally, the argument in Petitioner’s Reply that Marx would have motivated the Glu³ substitution because it teaches “the ionizable side chains of Asp², Asp³, and Glu⁵ ‘may be involved in the control and stabilization of the two’” topoisomers of human uroguanylin is not persuasive. Reply 13–14 (quoting Ex. 2010 (Marx), 236, 238). As Patent Owner points out, Marx

says only that the ionizable side chains of these residues “may be involved” and seemingly contradicts that observation with data it concludes show that “isomerization is not affected by the amino-terminal region of uroguanylin.” Sur-reply 11 (quoting Ex. 2010, 236) (emphasis omitted). Thus, we do not agree with Petitioner’s suggestion that Marx teaches that making the Glu³ substitution would affect interconversion between the topoisomers. Even if Marx were interpreted as Petitioner suggests, both Asp and Glu have an ionizable side chain. Making the Glu³ substitution would not change the fact that there is an ionizable side position. Thus, the record does support Petitioner’s suggestion that a POSA would have expected the Glu³ substitution to affect topoisomerization, much less that a POSA would have been motivated to make the Glu³ substitution for this reason.

iii. Unexpected results

Patent Owner offers evidence it contends shows that “the particular features of plecanatide” provide unexpected and superior results compared to human uroguanylin. *See* Resp. 58–66. Patent Owner presents this evidence in four categories of allegedly unexpected results: “(1) stabilization against interconversion, (2) unexpected superior potency, (3) surprisingly superior heat stability, and (4) unexpected superior binding affinity.” *Id.* at 58. According to Patent Owner, this evidence “underscores the lack of any reasonable expectation of success and supports the patentability of plecanatide.” *Id.*

Before turning to the specifics of Patent Owner’s evidence as it relates to each category of results, we address certain global issues with Patent Owner’s showing. Patent Owner primarily relies on tests and data from three

different sources: Table 4 in the Specification, Exhibit 2027, and Exhibit 2028.¹⁶ It is undisputed that the uroguanylin (also referred to as SP-301) in Patent Owner’s tests is human uroguanylin and that SP-304 is a peptide consisting of the amino acid sequence in SEQ. ID. NO: 20, as recited in claim 1. *See, e.g.*, Ex. 1001, 10:39–63 (providing definitions); Ex. 2027, TRUL00018210 (showing sequences), TRUL00018218 (correcting sequence for uroguanylin); Ex. 2024 ¶ 217; Ex. 2066 ¶ 13. The only difference in the amino acid sequence of SP-301 and SP-304 is the latter’s Glu³ substitution. *Id.* Patent Owner’s evidence also includes data for SP-302 and SP-303, which are other test peptides derived from human uroguanylin. SP-303 has the sequence of human uroguanylin, but with a Glu² substituted for the Asp². *Id.* SP-302 has both the Glu² and Glu³ substitution. *Id.*

Petitioner contends that the human uroguanylin (SP-301) in Patent Owner’s tests is not the closest prior art because it was tested as a mixture of the active and inactive topoisomers. *See* Reply 23. We disagree with Petitioner. As Patent Owner points out, “[h]uman uroguanylin naturally interconverts between, and exists as a mixture of, two isomers.” Sur-reply 20; *see also* Ex. 2011, 28 (“topological isomers of human uroguanylin are not stable in solution and are readily interconvertible”); Ex. 2024 ¶¶ 67–68; Ex. 2025 ¶¶ 49–52. The record also supports Patent Owner’s position that

¹⁶ Exhibits 2027 and 2028 are study reports submitted as part of Patent Owner’s New Drug Application (NDA) for a plecanatide-containing drug product. *See* Paper 26, 1–3; Paper 61, 9. These exhibits were initially filed under seal (*see id.*), but Patent Owner has since filed unredacted versions of those papers. While the unredacted exhibits still bear confidentiality designations, they are nevertheless publicly-available. *See* Paper 73 (Patent Owner agreeing to make these papers public).

“[s]ome amount of interconversion to the inactive conformation was inevitable” in its testing. Sur-reply 20 (citing Ex. 2010, 238; Ex. 2011, 31 (noting that “interconversion may occur during the course of the analysis” especially under acidic conditions)). Thus, Patent Owner’s tests reflect the use of human uroguanylin as it naturally exists, i.e., a mixture with some interconversion between topoisomers, which is the closest prior art to the peptide of claim 1.

Petitioner also contends that the alleged unexpected results are not reasonably commensurate with the scope of claim 1 because Patent Owner “never assert[s] their results represent all topoisomers of SEQ ID NO. 20” and claim 1 “neither requires a particular topoisomer nor excludes mixtures of SEQ ID NO. 20 topoisomers.” Reply 22–23. We disagree. First, the extent to which SP-304 interconverts between an active and inactive topoisomer, if at all, is not clear on this record. Petitioner and Dr. Peterson assert that Patent Owner’s “later patents . . . explain [that] the ’786 patent’s synthesis methods produce mixed topoisomers.” Reply 22; Ex. 1063 ¶ 148 (citing Ex. 1068, 3:30–41). But the exhibit they cite is more equivocal, stating only that “synthesis of peptides for pharmaceutical application present a number of special problems such as an overall low yield . . . and/or high levels of impurities (e.g., contaminants resulted from organic solvents used during syntheses or purification, and degradation products or topoisomers created, e.g., during purification).” Ex. 1068, 3:35–41. Moreover, as explained below, Patent Owner’s HPLC data show that, unlike human uroguanylin, SP-304 did not form an inactive topoisomer under the conditions tested there. *Infra* § III.E.iii.1. Thus, Petitioner has not

established the premise for its argument, i.e., that the peptide of claim 1 readily interconverts between an active *and* inactive topoisomer.

Second, even if it had been shown that the peptide of claim 1 is subject to topoisomerization, Patent Owner’s unexpected results evidence would still be reasonably commensurate with the scope of that claim. While “unexpected results must be commensurate in scope with the claims, “absolute identity of scope” is not required. *Genetics Institute, LLC v. Novartis Vaccines and Diagnostics, Inc.*, 655 F.3d 1291, 1308–09 (Fed. Cir. 2011) (citing cases rejecting unexpected results “where the evidence was plainly disproportionate to the scope of the claim”). Claim 1 here is narrowly drawn to a peptide consisting of a single 16 amino acid long sequence. All of the unexpected results Patent Owner alleges are based on data from tests of SP-304, i.e., a peptide consisting of that sequence. Thus, even if we assume there is more than one topoisomer of the sequence recited in claim 1 and the activity of those topoisomers differed from each other, Patent Owner’s SP-304 data would not be disproportionate to the overall scope of this narrowly-drawn claim.

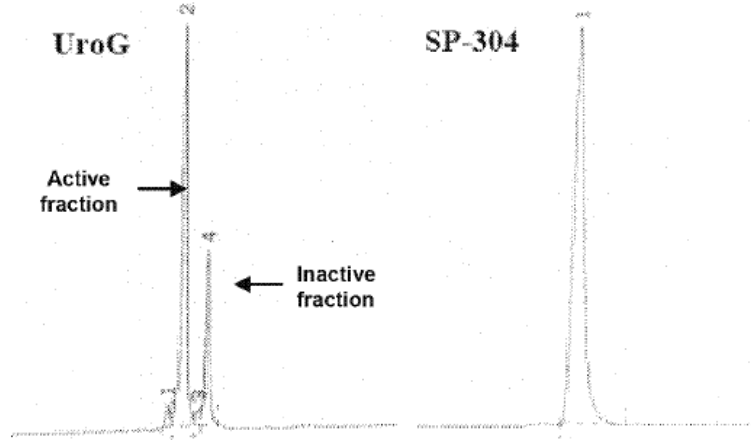
For these reasons, Patent Owner has established a sufficient nexus between claim 1 and its experimental results evidence and demonstrated that these results it describes compare the claimed peptide (SP-304) to the closest prior art. Having resolved these global issues, we turn to the specifics of Patent Owner’s evidence for each of its categories of results.

1. Interconversion stability

Patent Owner argues that “[p]lecanatide’s particular combination of structural features yielded a peptide that is unexpectedly stabilized against

interconversion as compared to human uroguanylin.” Resp. 58. As evidence, Patent Owner offers data from an experiment involving “[s]amples of plecanatide (SP-304) and human uroguanylin (UroG) [that] were incubated for 16 hours at 37°C in a solution having a pH of 3.0” and “then analyzed by HPLC [i.e., high pressure liquid chromatography].” *Id.* (citing Ex. 2028, TRUL00018269). The results of that analysis are shown in the figure reproduced below. Ex. 2028, TRUL00018269.

Figure 1: HPLC Chromatographs of Uroguanylin and SP-304 Following 16-Hour Incubations at pH 3.0 in Aqueous Media at 37°C



The figure above shows HPLC chromatographs of uroguanylin on the left and SP-304 on the right. The uroguanylin chromatograph shows a first peak labeled “Active fraction” next to a second, smaller peak labeled “Inactive fraction.” *Id.* In contrast, the SP-304 chromatograph shows only a single peak. According to Patent Owner, the result shown in this data “would not have been expected, for nothing in the art would have suggested that the Asp³ to Glu³ substitution would have improved human uroguanylin’s isomeric stability, let alone to this degree.” Resp. 59 (citing Ex. 2024 ¶¶ 210–215).

We find Patent Owner’s evidence persuasive to show that the Glu³ substitution results in an unexpected improvement in interconversion stability. HPLC was a known technique for separating the topoisomers of human uroguanylin. Ex. 2010, 230–231; Ex. 2011, 27–28; Ex. 2020, 223; Ex. 2028, TRUL00018269; *see also* Ex. 1063 ¶ 40 (Petitioner’s expert confirming same). It was also known that one of human uroguanylin’s topoisomers was biologically active, whereas the other was “practically inactive.” Ex. 2011, 31; *see also* Ex. 2010, 229 (noting “the existence of two topological stereoisomers of which only one induces intracellular cGMP elevation”). The chromatograph for uroguanylin in Patent Owner’s data is consistent with these teachings, showing two peaks representing the active and inactive topoisomers. *Compare* Ex. 2028, TRUL00018269 (Fig. 1) *with* Ex. 2011, 28 (Fig. 2). The chromatograph for SP-304 shows a markedly different result, i.e., a “single peak,” which was confirmed “using the cGMPT T84 cell assay” to represent “the fully active conformer of SP-304.” Ex. 2028, TRUL00018269; Ex. 2024 ¶¶ 208–209.

We agree with Patent Owner and credit Dr. Davies’s testimony (Ex. 2024 ¶¶ 210–215) that a POSA would not have expected the Glu³ substitution to produce such a result. Indeed, the premise for Petitioner’s “conservative substitution” rationale is that the Glu³ substitution would have been expected to produce an analog having similar properties to human uroguanylin. *See, e.g.*, Pet. 1, 36 (stating a POSA “would have expected this particular conservative substitution to retain excellent receptor-activating activity”); Reply 3 (urging that the reason to modify a known compound may be an expectation that the new compound will have “similar properties to the old”) (internal quotations omitted). Patent Owner’s HPLC data,

however, demonstrate a difference in kind between the data for SP-304 (a single HPLC peak) and uroguanylin (two distinct peaks corresponding to the active and inactive topoisomer) after incubation under the same conditions. As Dr. Davies explains, “topoisomers exist as a mixture of compounds with different[] spatial conformations” that are “continuously interconverting between spatial conformations” with “varying degrees of interconversion stability.” See Ex. 2024 ¶¶ 51–56. Thus, Patent Owner’s data show that the Glu³ substitution results in a peptide that, assuming it interconverts at all, is more stable against interconversion than the parent, unmodified peptide.

Marx further shows that this result would not have been expected. The Asp³ residue in human uroguanylin is the third residue from the amino-terminal end of that peptide. But Marx teaches that “isomerization is not affected by the amino-terminal region of uroguanylin.” Ex. 2010, 236. Marx also suggests that a different modification, i.e., adding a “third disulfide bond apparently would lead to a preference of a structure similar to the” biologically active isomer. See Resp. 39–40 (quoting Ex. 2010, 235). Thus, Marx suggests that a POSA seeking to stabilize uroguanylin against interconversion to the inactive isomer should pursue modifications other than the Glu³ substitution.

Nevertheless, Petitioner cites Marx and Klodt, urging that the “prior art suggested acidic residues at positions 2-3 contribute to stabilization.” Reply 28. We disagree. As explained above, Marx teaches that it is the “ionizable side-chains” in Asp² and Asp³ that “may be involved in the control of stabilization of the two isomers.” Ex. 2010, 236. The substitution of one acidic residue (Asp) with another residue having an acidic side chain (Glu) does not affect this; both the Glu³ and Asp³ peptide would still have an

ionizable side chain at the same position. Petitioner also refers to Klodt's observation that "the 'sterical bulk' of 'other residues' far from the C-terminus may influence interconversion." Reply 13 (quoting Ex. 2020, 228). But that teaching does not identify Asp³. Rather, what Klodt says is that "other residues, *in particular between the inner cysteine residues*, also influence [stability of topoisomers] by their sterial bulk." Ex. 2020, 228 (emphasis added). Asp³ is not located between the cysteine residues in human uroguanylin. *See* Ex. 1005, 1:50–55 (SEQ ID NO: 1); Ex. 1006, 52 (Fig. 6a) (showing that human uroguanylin has cysteine residues at positions 4, 7, 12, and 15). Accordingly, neither Marx nor Klodt supports Petitioner's position that a POSA would have expected the result shown in Patent Owner's HPLC data.

Petitioner's argument that Patent Owner has not shown that the incubation conditions for its experiment "(e.g., pH 3 for 16 hours)" are "relevant" is unavailing. Reply 27. First, Petitioner does not explain what it objects to regarding these incubation conditions. *See id.* To the extent Petitioner contends the conditions differ from those *in vivo* when a peptide is orally administered, that distinction does not undermine Patent Owner's showing because the data still show that plecanatide exhibited unexpectedly improved interconversion stability under the conditions tested. Moreover, Petitioner's suggestion that these conditions are somehow not relevant is at odds with the fact that the data were apparently submitted to the FDA as part of Patent Owner's NDA. *See* Paper 61, 9 (Patent Owner representing that "Preclinical Study Report Nos. SP-PH-001 (Ex. 2027) and SP-PH-004 (Ex. 2028) are a part of the NDA submitted to the FDA for Trulance®"); Paper 26, 1 (similar).

Petitioner’s argument that Patent Owner should have additionally provided NMR data to confirm the structure of the peptide in each HPLC peak because “its HPLC protocol may have been optimized to separate topoisomers of [uroguanylin] but not SP-304” is not persuasive. Reply 28 (citing Ex. 1063 ¶¶ 146–147). HPLC was a known technique for separating topoisomers of uroguanylin. Ex. 2010, 230–31; Ex. 2011, 27–28; Ex. 2020, 223; Ex. 2028, TRUL00018269; Ex. 1063 ¶ 40. Thus, we see no reason why Patent Owner’s HPLC data are not itself sufficient to show the alleged results. Indeed, the premise for Patent Owner’s argument, i.e., Dr. Peterson’s testimony that the SP-304 peak “is relatively broad with some degree of tailing, raising the question of whether the HPLC protocol employed was simply lacking in optimization to resolve the topoisomers of this peptide” (Ex. 1063 ¶¶ 146–147), is just speculation. In contrast, we find the testimony Patent Owner offers from Dr. Shailubhai—who “directed the preparation and testing” resulting in Patent Owner’s data, provides detailed information regarding the HPLC conditions, and concludes that the chromatograph shows “only a single peak” for SP-304 (Ex. 2066 ¶¶ 17–20)—more credible than the speculative testimony from Dr. Peterson that Petitioner cites.

2. Potency

Patent Owner argues that “Plecanatide is also unexpectedly more potent than human uroguanylin, stimulating significantly more cGMP production in T84 cells.” Resp. 60. In addition to Table 4 in the Specification, Patent Owner relies on the following data from Table 1 in Exhibit 2027. *Id.* at 61 (reproducing Ex. 2027, TRUL00018222 (Table 1)); *see also* Ex. 1001, 16:1–18 (Table 4) (showing some of the same results).

Table 1. Effects of SP-304, Uroguanylin, and Other Test Peptides in the T84 cGMP Stimulation Bioassay

Test Material	Concentration		cGMP Levels (pmol/well) *	EC ₅₀ **	
	Molar	ng/mL		Molar	ng/mL
SP-301 (uroguanylin)	0	0	0	10 ⁻⁶ M 2.3x10 ⁻⁷ M	1668 383.6
	10 ⁻⁹ M	1.668	0		
	10 ⁻⁸ M	16.668	12		
	10 ⁻⁷ M	166.8	82		
	10 ⁻⁶ M	1668	205		
	10 ⁻⁵ M	16680	254		
SP-302	0	0	0	10 ⁻⁶ M 3.5x10 ⁻⁷ M	1696 593.6
	10 ⁻⁹ M	1.696	0		
	10 ⁻⁸ M	16.96	8		
	10 ⁻⁷ M	169.6	62		
	10 ⁻⁶ M	1696	185		
	10 ⁻⁵ M	16960	248		
SP-303	0	0	0	10 ⁻⁶ M 2.4x10 ⁻⁷ M	1682 403.7
	10 ⁻⁹ M	1.682	0		
	10 ⁻⁸ M	16.82	12		
	10 ⁻⁷ M	168.2	82		
	10 ⁻⁶ M	1682	195		
	10 ⁻⁵ M	16820	254		
SP-304	0	0	0	10 ⁻⁷ M 1.1x10 ⁻⁷ M	168.2 185.0
	10 ⁻⁹ M	1.682	0		
	10 ⁻⁸ M	16.82	17		
	10 ⁻⁷ M	168.2	149		
	10 ⁻⁶ M	1682	320		
	10 ⁻⁵ M	16820	315		

* cGMP levels in T84 cells after a 30-minute incubation.

** EC₅₀: median effective concentration (required to induce a 50% effect)

Table 1 above shows data for plecanatide (SP-304), human uroguanylin (SP-301) and other test peptides (SP-302 and SP-303) at different concentrations in the T84 cGMP stimulation bioassay. According to Patent Owner, these data show that “plecanatide increased cGMP production by about 56% at 10⁻⁶ M (320 v. 205 pmol/well)” and “achieved 50% of its maximal effect at less than half of the concentration of human uroguanylin (1.x10⁻⁷ M v. 2.3x10⁻⁷M).” *Id.* at 60.

Patent Owner also relies on the data in Figure 2 of Exhibit 2028, reproduced below, showing potency of human uroguanylin, SP-302, SP-303, and SP-304 at pH 5.5, 6.0, 6.5, and 7.0. Resp. 61–62 (citing Ex. 2028, TRUL00018270); *see also* Ex. 2028, TRUL00018273 (showing raw data).

Figure 2: Effect of 30 Minute Incubations of 0.1 μ M Concentrations of SP-304, Uroguanylin, SP-302 and SP-303 on cGMP Production at Various Physiological pH Values in T84 Cells

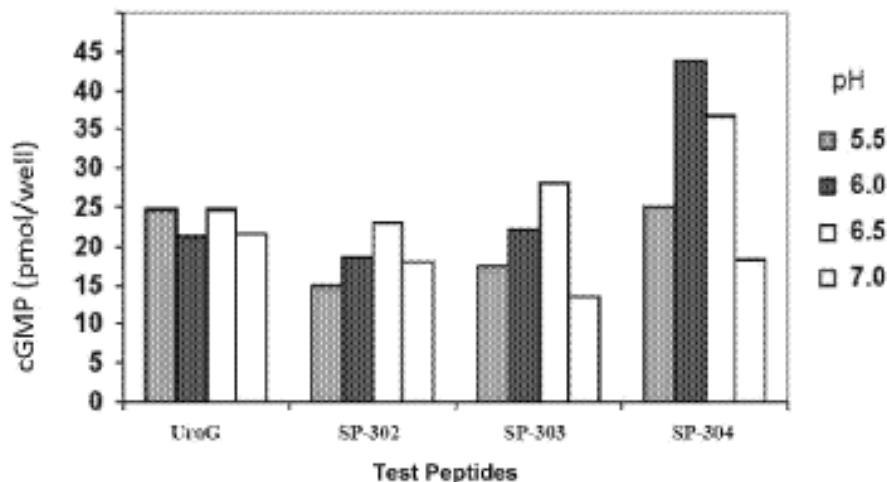


Figure 2 above is a bar graph of cGMP data for plecanatide, human uroguanylin, SP-302, and SP-303 at different pHs ranging from 5.5 to 7.0. According to Patent Owner, these data show “as compared to human uroguanylin, plecanatide increased cGMP production by about 105% at pH 6.0 and by about 49% at pH 6.5.” Resp. 61–62. Relying on testimony from Dr. Waldman, Patent Owner argues that this “increased pH sensitivity and better potency in the acidic portion of the intestine allows higher fluid secretion to treat constipation while its activity advantageously drops at pH 7 (in the colon) to minimize the likelihood of diarrhea.” *Id.* at 63 (citing Ex. 2025 ¶¶ 103–104, 110, 111).

Petitioner replies that the “magnitude” of the differences in Patent Owner’s potency data “is well within the . . . experimental error for these experiments and does not demonstrate any real, significant, or material difference in cGMP activity between the two peptides.” Reply 24 (citing Ex. 1063 ¶¶ 118–135). Referring to Table 4 in the Specification, Petitioner urges that its “cGMP values cannot be read to conclude a statistically

significant difference exists between the tested peptides, as even Dr. Shailubhai confirmed.” *Id.* (citing Ex. 1061, 33:19–34:15). Petitioner also asserts that “Dr. Davies testified a 40% difference in cGMP values for the same peptide in the same experiment does not reflect an actual difference but merely ‘a different way to interpret the data.’” *Id.* at 24–25 (quoting and citing Ex. 1060, 57:8–10, 54:3–4, 61:8–18, 74:8–17). Petitioner further urges that the raw data in Tables 2–3 of Exhibit 2028 “show **higher** nominal cGMP values **for uroguanylin** than for plecanatide at the same concentration with the same assay protocol.” *Id.* at 25 (citing Ex. 1063 ¶¶ 131–134). Moreover, Petitioner points to other data Patent Owner submitted to FDA, which it contends “indicates both peptides induced similar cGMP-production levels at the same concentration used to generate the Table 1 data [in Exhibit 2027] that was included in Table 4 [of the Specification].” *Id.* at 26 (citing Ex. 1069, 26, 30).

We find Patent Owner’s showing of unexpectedly improved potency and pH sensitivity persuasive. Patent Owner provides tests showing “the stimulation of cGMP synthesis in T84 cells by plecanatide (SP-304), human uroguanylin (SP-301)” as well as SP-302 and SP-303 as a measure of these peptides’ relative potency. *See* Ex. 2024 ¶¶ 217, 222, 224. Both the data in Table 4 of the Specification and Table 1 of Exhibit 2027 show that SP-304 produced a higher cGMP level in the T84 cell bioassay than SP-301 (uroguanylin). *See* Ex. 1001, 16:1–18 (Table 4); Ex. 2027, TRUL00018222 (Table 1). As Patent Owner points out, the difference at 10^{-6} M was more than 50% higher. *Id.* Moreover, Table 1 shows that SP-304 produced higher cGMP over a wide range of tested concentrations (10^{-7} M to 10^{-5} M) and that SP-304’s EC_{50} , i.e., the “median effective concentration (required to induce

a 50% effect),” was about half that of uroguanylin.¹⁷ Ex. 2027, TRUL00018222 (Table 1). The data also show that SP-302 and SP-303 produced values similar to or slightly worse than uroguanylin at 10^{-6} M. *See* Ex. 1001, 16:1–18 (Table 4); Ex. 2027, TRUL00018222 (Table 1). Thus, Patent Owner’s data show that SP-304 exhibited improved potency values compared to both the parent peptide, human uroguanylin, as well as two other test peptides (SP-302 and SP-303) comprising different substitutions of the aspartic acid residues at positions 2 and 3.

The data in Figure 2 of Exhibit 2028 also supports Patent Owner’s position. As Patent Owner points out, the data show that the Glu³ substitution resulted in increased cGMP production at pH 6 and 6.5 as compared to uroguanylin and the other tested peptides. Resp. 61–62 (citing Ex. 2028, TRUL00018270, TRUL000018273). Dr. Waldman offers testimony, which we credit, that both the increased activity and “pH sensitivity” are unexpected and “clinically meaningful” because

this unexpected pH sensitivity in the acidic areas of the intestines allows plecanatide’s unexpectedly superior selective activity in the [pH environment of] the small intestine to provide fluid into the intestine to treat constipation, rather than in the colon, where its activity drops (pH 7) and is thus less likely to cause adverse effects, like diarrhea. In other words, plecanatide’s targeted activity in the small intestine (not in the

¹⁷ Petitioner asserts that the EC₅₀ values in Table 1 were amended to “eliminate[] the purported 10-fold potency difference” shown in the original version of that table. Reply 24. While this is true, Petitioner’s observation is inapposite because Patent Owner relies only on the amended values. *See* Resp. 60–61 (showing Table 1 as amended). Moreover, as Patent Owner explains, this amendment did not change the underlying cGMP data used to calculate those EC₅₀ values. Sur-reply (citing Ex. 2027, TRUL00018222).

colon) advantageously reduces the diarrhea rate in patient populations.

Ex. 2025 ¶ 110; *see also* Ex. 2025 ¶¶ 37–38 (explaining that “[a]s of 2002, ***all*** available treatment options [for constipation] targeted the colon or both the colon and small intestine,” which “is in stark contrast to plecanatide, which even today, is the only [chronic idiopathic constipation] and [irritable bowel syndrome with constipation] treatment to preferentially target the small intestine”). These differences, particularly the improved pH sensitivity at pH 6.0 and 6.5 shown in Figure 2 of Exhibit 2028, represent a difference in kind from the data for human uroguanylin.

Dr. Davies’s testimony that these results would have been unexpected because they show “that similar and/or seemingly minor modifications to human uroguanylin result in significant changes to biological activity in unpredictable ways” is persuasive. *See* Ex. 2024 ¶¶ 228–230. Li identifies both of the aspartic acid residues at position 2 and 3 in human uroguanylin as being of “particular interest.” *See* Ex. 1006, 52 (Fig. 6a), 54. And Petitioner appears to acknowledge that according to its own logic, a POSA relying on Li would also have been motivated to make the Glu² substitution. *See* Pet. 39 (“a POSA could also make [Glu²]-human uroguanylin as an operable synthetic uroguanylin”). However, Patent Owner’s data show that surprisingly *only* the Glu³ substitution results in improved potency compared to uroguanylin, whereas other conservative substitutions (i.e., the Glu² in SP-303 and the combination of the Glu² and Glu³ substitutions in SP-302) resulted in peptides with cGMP and EC₅₀ values similar to or worse than the parent peptide. Nothing in the record suggests such results would have been expected.

Petitioner's argument that the differences between the values in Patent Owner's data are immaterial and can be attributed to experimental error is unpersuasive. The data Patent Owner cites consistently show higher activity for SP-304 as compared to the other peptides tested. Moreover, Table 4 in the Specification states that the result for SP-304 is "statistically significant." Ex. 1001, 16:19.¹⁸ Petitioner asserts that Dr. Shailubhai "confirmed" that result was not statistically significant in his deposition. Reply 24 (citing Ex. 1061, 33:19–34:15). But all Dr. Shailubhai said was that the P value was calculated by someone else on his team and that he could not recall what SP-304 was compared to for that calculation. Ex. 1061, 33:19–34:15. None of that suggests the SP-304 result in Table 4 is not, in fact, significant.

So too, the testimony Petitioner cites from its cross-examination of Dr. Davies is inapposite and does not undermine Patent Owner's showing. *See id.* at 24–25 (quoting and citing Ex. 1060, 57:8–10, 54:3–4, 61:8–18, 74:8–17). That testimony relates to a discrepancy between the value reported for SP-302 in Table 4 (225 pmol/well) and the value reported for SP-302 at a 10^{-6} M concentration in Table 1 of Exhibit 2027 (185 pmol/well). Ex. 1060, 70:6–71:5. Both SP-302 values are still well below the cGMP value reported for SP-304 (315 pmol/well) and we credit Dr. Davies's testimony that looking at the data "as whole, it doesn't change any interpretation you would put on the data." *Id.* at 74:8–22.

¹⁸ The note below Table 4 states "[t]he value observed for SP304 was statistically significant with a $p > 0.5$." Ex. 1001, 16:19. Dr. Shailubhai explains—and Petitioner does not dispute—that there is a typographical error with the p value and that it "should say $p < 0.05$." Ex. 2066 ¶ 9.

Dr. Peterson's testimony that a difference in the results for SP-304 in Patent Owner's data can be attributed to experimental error is not persuasive. *See* Ex. 1063 ¶¶ 118–35. Patent Owner offers multiple data sets, showing improved potency values for SP-304 relative to uroguanylin, SP-302, and SP-303. Ex. 1001, 16:1–18 (Table 4); Ex. 2027, TRUL00018222 (Table 1); Ex. 2028, TRUL00018270, TRUL00018273 (Table 2). In response, Dr. Peterson suggests that other statistical analysis could have been done or that these experiments should have been replicated to confirm their reliability. *See* Ex. 1063 ¶¶ 122–23. But that does not undermine the veracity of the data Patent Owner has presented.¹⁹ Dr. Peterson also cites the same deposition testimony of Dr. Shailubhai and Dr. Davies discussed above, picking at the edges of Patent Owner's evidence by comparing results from different experiments in an attempt to suggest that any difference between the values for uroguanylin and SP-304 is unreliable. *See* Ex. 1063 ¶¶ 125–135. For example, Petitioner and Dr. Peterson point to the raw data for Patent Owner's pH sensitivity assays in Table 2 and compare them to the data for the thermostability assays in Table 3 of Exhibit 2028. *See* Reply 25 (citing Ex. 1063 ¶¶ 131–34; Ex. 2028, TRUL00018273 (Tables 2–3)). But these are different experiments and Petitioner has not sufficiently shown that the conditions (e.g., pH) are similar such that the comparison Dr. Peterson makes is accurate. For these reasons, we credit the testimony of Dr. Davies and Dr. Waldman that Patent Owner's data demonstrate significant and unexpectedly improved potency and pH sensitivity for SP-304 as compared

¹⁹ Moreover, at least for the results in Table 1 the “[s]amples were run in duplicate[] incubations and each sample was run as duplicates in ELISA test.” Ex. 2027, TRUL00018211; Ex. 2023 ¶ 15.

to uroguanylin (see Ex. 2024 ¶¶ 216–30; Ex. 2025 ¶¶ 95–111) over the competing testimony Petitioner offers from its experts.

3. Heat stability

Patent Owner asserts that “[p]lecanatide is also unexpectedly more heat stable than human uroguanylin.” Resp. 64. Patent Owner relies on data from an experiment in which SP-304 and human uroguanylin “were compared after heating at 95°C for up to 90 minutes” and then tested “[i]n the T84 cell-based assay at specific times.” *Id.* (citing Ex. 2028, TRUL00018268, TRUL00018270–71). Data for this experiment are depicted in Figure 3 of Exhibit 2028, which is reproduced below.

Figure 3: Thermostability of SP-304, Uroguanylin, SP-302 and SP-303 Following Heat Treatment at 95°C in T84 Cells as a Function of Time

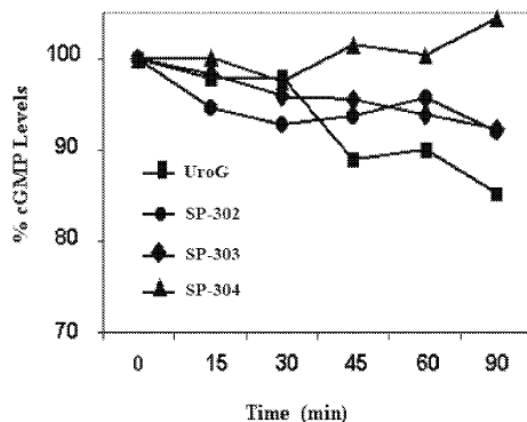


Figure 3 above is a line graph showing the change in “% cGMP Levels” of plecanatide, human uroguanylin, SP-302, and SP-303 over time. Ex. 2028, TRUL00018271. According to Patent Owner, the data in Figure 3 show that SP-304 retained virtually all of its activity, whereas “uroguanylin lost about 15% of its activity after 90 minutes.” *Id.* (quoting Ex. 2028, TRUL00018270). Patent Owner argues this result “would have been unexpected in view of the one methylene group difference” between

glutamic acid and aspartic acid and that “[a] POSA would have expected, at best, that plecanatide would have had a similar thermostability to human uroguanylin, as was reported for peptides with other minor variations like SP-302 and SP-303.” *Id.* at 65 (citing Ex. 2024 ¶¶ 236, 238).

Petitioner replies that “the data do not support [Patent Owner’s] argument.” Reply 26. According to Petitioner, Patent Owner and Dr. Davies “overstate the difference between the nominal cGMP value observed for SP-301 after heat treatment (10–13%, not 15% or 20%), ignore that SP-301 and SP-303’s initial cGMP values were 6-13% greater than those for SP-304, and present the data to create a false impression that significant or material heat-stability differences exist.” *Id.* at 26–27 (citing Ex. 1063 ¶¶ 136–138, 141–143). In addition, Petitioner argues that the “nominal cGMP values” in Patent Owner’s data “repeatedly swing back and forth” and that “SP-304’s cGMP values begin much lower than the other peptides and end ~7% **higher after** heat treatment, and even the patent admits ‘[b]iological activity . . . did not change significantly.’” *Id.* at 27 (quoting Ex. 1001, 16:19–28). For these reasons, Petitioner urges that Patent Owner “fails to demonstrate any material difference in kind between peptides” with respect to their thermostability. *Id.* (citing Ex. 1063 ¶¶ 135–144, 176).

We agree with Petitioner that Patent Owner’s thermostability data do sufficiently demonstrate a difference in kind between SP-304 and human uroguanylin. The raw data for Figure 3 is shown in Table 3 below. Ex. 2028, TRUL00018273.

Table 3: Raw Data (thermostability assays)

Test Peptide	Concentration	cGMP Levels (pmol/well) *					
		0	15 min	30 min	45 min	60 min	90 min
Uroguanylin	0.1 μ M (1.67 μ g/mL)	21.72	20.92	21.68	20.88	18.76	18.96
SP-304	0.1 μ M (16.8 μ g/mL)	20.44	21.12	20.64	21.4	20.8	21.88
SP-302	0.1 μ M (1.67 μ g/mL)	23.16	20.44	21.12	20.32	20.36	20.34
SP-303	0.1 μ M (1.68 μ g/mL)	21.64	20.0	20.56	20.68	20.36	19.8

* cGMP levels in T84 cells after heat treatment at 95°C for the indicated times. Samples at different times of treatment were withdrawn and assayed for their ability to stimulate cGMP synthesis.

The data in Table 3 show that cGMP levels for SP-304 vacillated at the different time points in the experiment, ranging from as low as 20.44 pmol/well to as high as 21.88 pmol/well after heat treatment for 90 minutes. *Id.* As Dr. Peterson explains, this is a nominal increase of about 7% that “cannot be attributed to reversing degradation of the peptide” through heat treatment, which, if anything, would reduce activity. Ex. 1063 ¶ 139. Thus, this increase is “attribut[able] to experimental error.” *Id.*

Similar vacillation can be seen between some of the data points for the other peptides in Table 3 with the nominal cGMP level increasing slightly notwithstanding additional heat treatment. In particular, uroguanylin’s cGMP level after 60 minutes (18.76 pmol/well) is slightly lower than its activity after 90 minutes (18.96 pmol/well). *See also* Ex. 2028, TRUL00018271 (Figure 3 showing uroguanylin had lost about 10% of its original cGMP level after 60 minutes). The result in Exhibit 2028 is contradicted by the Specification, which describes results from what appears to be a similar thermostability assay,²⁰ but states that the “[b]iological

²⁰ While the description of the protocols is similar, it is unclear if the Specification describes the same or merely a very similar experiment to that described in Exhibit 2028. *See* Ex. 1060, 47:3–22 (Dr. Davies testifying that “[i]t certainly looks like the same experiment”). In either event, the

activity of [uroguanylin] did not change significantly after 60 minutes” and was “reduced to about 80%” of its original value after 90 minutes. Ex. 1001, 16:23–28.

Neither Patent Owner, nor Dr. Davies, sufficiently account for these discrepancies. See Sur-reply 25 (referring to “downward trajectory” of uroguanylin’s heat stability compared to plecanatide). Dr. Davies charts the data from Table 3 using a linear regression model to try to show a difference in the “line trends” between uroguanylin and SP-304. Ex. 2024 ¶¶ 235–36. But that analysis does not explain the discrepancy between the line trend for uroguanylin and the Specification’s observation that activity for uroguanylin “did not change significantly after 60 minutes.” Ex. 1001, 16:23–28. Moreover, unlike the data Patent Owner relies upon for its other unexpected results arguments, the magnitude of the purported differences between the values for SP-304 and uroguanylin in Table 3 of Exhibit 2028 are much smaller. For these reasons, we agree with Petitioner that Patent Owner’s thermostability data does not sufficiently demonstrate a difference in kind between the thermostability of SP-304 compared to that of human uroguanylin.

4. Binding affinity

Patent Owner asserts that “[p]lecanatide also has an unexpected and superior binding affinity as compared to human uroguanylin.” Resp. 66. Patent Owner relies on data from Liu (Exhibit 2046), a research article published in the Anticancer Research journal in 2009. Ex. 2046, 3777.

description of the results in the Specification is inconsistent with the data in Table 3 of Exhibit 2028.

According to Patent Owner, “Liu conducted an independent study demonstrating, and as shown in Table 1 [reproduced below], plecanatide (E3-uroguanylin) binds to the GCC receptor with **8-fold higher affinity** than native human uroguanylin.” Resp. 66 (reproducing table from Ex. 2046, 3779).

Table I. Calculated and observed (M+H)⁺ values and IC₅₀ values (± SD) for characterized peptides.

Peptide	(M+H) ⁺ Calc.	(M+H) ⁺ Obs.	IC ₅₀ (nM)
Uroguanylin	1667.6	1667.7	39.8±14.9
DOTA-uroguanylin	2053.6	2053.9	34.5±3.3
E ³ -uroguanylin	1681.6	1681.6	5.0±0.3
DOTA-E ³ -uroguanylin	2067.6	2067.9	9.6±2.9

Table 1 above shows IC₅₀ values²¹ from a competitive binding assay described in Liu. See Ex. 2046, 3778–3779. Patent Owner compares the IC₅₀ value reported for uroguanylin (39.8 ± 14.9 nM) to the IC₅₀ value for E³-uroguanylin (5.0 ± 0.3 nM) from Table 1 above. Relying on testimony from Dr. Davies and Dr. Waldman, Patent Owner argues this approximately “8-fold increase in binding affinity would have been unexpected given the single methylene group difference [between aspartic acid and glutamic acid] and also meaningful, for it demonstrates substantially greater potency for cGMP production to treat constipation as shown above.” Resp. 66 (citing Ex. 2024 ¶ 243; Ex. 2025 ¶¶ 94, 110–11).

²¹ As Dr. Davies explains, “IC₅₀ is a quantitative measure that indicates how much (or what concentration) of a particular inhibitory substance (e.g. drug) is needed to inhibit, in vitro, a given biological process or biological component by 50%.” Ex. 2024 ¶ 239; see also Ex. 2025 ¶ 90 (similar explanation from Dr. Waldman).

Petitioner does not dispute that the data in Liu's Table 1 show an 8-fold increase in binding affinity. *See* Reply 26. Instead, Petitioner urges this is not an unexpected result because Patent Owner "does not contend this binding-affinity increase translated into an **8-fold** potency increase." *Id.* (citing Ex. 1064 ¶¶ 47–49).

We find Patent Owner's showing of unexpectedly improved binding affinity persuasive. The data in Liu's Table 1 shows that plecanatide exhibited significantly improved binding affinity over uroguanylin. Indeed, Liu observes that "[a] conservative D3E substitution [i.e., the Glu³ substitution] significantly increased the binding affinity of the resultant uroguanylin analogs." Ex. 2046, 3779. This observation supports the testimony of Patent Owner's experts, which we credit, that a POSA would not have expected this conservative substitution to result in such a large improvement in binding affinity. Ex. 2024 ¶ 243; Ex. 2025 ¶ 94.

The record also shows that binding affinity is a relevant consideration because it is an indicator of potency. Dr. Davies testifies that plecanatide's "unexpectedly better" IC₅₀ "is indicative of the superior activity of plecanatide." Ex. 2024 ¶¶ 241–43. Dr. Waldman similarly testifies that this result is "significant because plecanatide would be seen as more potent in activating the receptors to stimulate greater cGMP production and potentially better relief of constipation." Ex. 2025 ¶ 94. Their testimony is consistent with the use of binding affinity assays to analyze and compare similar peptides in various other references. *See, e.g.*, Ex. 1005, Fig. 3B, 6:15–22; Ex. 1021, 2709; Ex. 2065, F345F (using binding assays to analyze biological activity and relative potency of uroguanylin and guanylin).

Petitioner’s argument that Patent Owner’s showing is insufficient unless the difference in binding affinity translates into a corresponding potency increase is not credible given Petitioner’s own reliance on such data. *See* Reply 26. As Patent Owner points out, Currie uses a similar competitive binding assay to assess the “relative affinity” of human uroguanylin, human guanylin, and STa. *See* Sur-reply 25; Ex. 1005, Fig. 3B, 6:15–22. Petitioner relies on this data, reproducing the binding affinity results shown in Currie’s Figure 3B in the Petition and urging that “Figure 3B . . . illustrates the increased potency of human uroguanylin compared to human guanylin.” Pet. 23. So too, Li cites uroguanylin’s “10-fold higher . . . affinity” as support for its teaching identifying certain positions of “particular interest.” *See* Ex. 1006, 54. As explained above, this is the same teaching Petitioner relies upon for its conservative substitution argument. *See supra* § III.E.ii.

The testimony Petitioner cites from Dr. Epstein does not show otherwise. According to Dr. Epstein, the result in Liu “would not have been viewed as ‘superior’ or ‘unexpected’ from a clinical standpoint” because “[b]inding affinity is really only relevant to a clinician to the extent it translates to activity.” Ex. 1064 ¶ 48. To the extent Dr. Epstein is suggesting that a POSA would not consider significantly improved binding affinity relevant, his testimony is belied by Currie, Li, and the other evidence noted above. Accordingly, we do not find the testimony Petitioner cites (i.e., Ex. 1064 ¶¶ 47–49) credible to rebut Patent Owner’s showing of significant and unexpectedly improved binding affinity.

iv. Conclusion for Ground 1

Weighing the record as a whole in view of the findings and analysis above, we conclude that Petitioner has not met its burden to show that claim 1 would have been obvious over Currie and Li. The evidence suggesting that it would have been obvious to make the Glu³ substitution because it is a conservative substitution is substantially outweighed by Patent Owner's comparatively strong evidence demonstrating unexpected results relating to improved interconversion stability, potency, and binding affinity from the Glu³ substitution. Indeed, Patent Owner's unexpected results showing is sufficiently strong that we would reach the same conclusion even if we had found the other rationales Petitioner offers for making the Glu³ substitution (i.e., higher side-chain pKa and avoiding aspartimide formation) to be persuasive. *See Millennium Pharms., Inc. v. Sandoz Inc.*, 862 F.3d 1356, 1368 (Fed. Cir. 2017) ("Evidence of objective indicia [such as unexpected results] 'can be the most probative evidence of nonobviousness in the record,' and . . . enable[s] 'the court to avert the trap of hindsight.'") (quoting *Leo Pharm. Prods., Ltd. v. Rea*, 726 F.3d 1346, 1358 (Fed. Cir. 2013)). Accordingly, Petitioner has not shown by a preponderance of the evidence that claim 1 is unpatentable.

F. Other Grounds

Petitioner's other grounds challenge dependent claims 2–6 and are premised on Petitioner's showing in Ground 1 for the limitations of claim 1. *See* Pet. 40–53. Petitioner relies on the additional references asserted in these grounds only for their teachings regarding the additional limitations recited in the dependent claims. *Id.* Because Petitioner has not shown that claim 1

would have been obvious over Currie and Li, the premise for Petitioner's other grounds has not been established. Accordingly, Petitioner has not shown by a preponderance of the evidence that claims 2–6 are unpatentable.

IV. PATENT OWNER'S MOTION TO EXCLUDE

Patent Owner moves to exclude Exhibit 1067 and paragraphs 34, 149, 152–157, 160, 162–164, and 166–167 of Exhibit 1063, which are portions of Dr. Peterson's second declaration relying on Exhibit 1067. PO MTE 1. Exhibit 1067 is "an excerpt from the file history of European Patent (EP) No. 1 379 224," the European analogue to the '786 patent, which was previously the subject of an opposition proceeding before the European Patent Office (EPO). *See id.* As Patent Owner explains, Exhibit 1067 "contains statements including data, made by the opponent [i.e., a party opposing the EP patent in the opposition proceeding] in Europe and its witness, Mark Currie." *Id.* at 1; *see also* Ex. 1067, 88–98, 120–127 (letter briefs of "Opponent 1" and "Annex A: Experimental Report" to those letter briefs describing "experiments . . . performed on behalf of Opponent 1").

Patent Owner argues that Exhibit 1067 should be excluded because the statements Petitioner relies upon in that exhibit are inadmissible hearsay under Federal Rule of Evidence (FRE) 802 and the data therein is not offered in accordance with 37 C.F.R. § 42.65(b), which requires that a party who relies on a technical test or data from such a test provide an affidavit with certain information. *Id.* at 1–5.²² Patent Owner raises similar arguments

²² Patent Owner also seeks to exclude Ex. 1067 under FRE 106 "because it is an incomplete excerpt of the file history." PO MTE 5. We do not agree that Exhibit 1067 should be excluded under FRE 106. As Petitioner points out,

for excluding the testimony of Dr. Peterson that relies on Exhibit 1067. *See id.* at 5–9 (arguing the testimony should be excluded under FRE 402, 403, and 802 because it is premised on inadmissible experiments and data described in Exhibit 1067).

Petitioner opposes Patent Owner’s motion to exclude. Paper 61 (“Pet. Opp.”). According to Petitioner, Exhibit 1067 is an admissible record of a public office that falls under the hearsay exception in FRE 803(8). Pet. Opp. 1–2. Petitioner also argues that Exhibit 1067 is not hearsay because it is “offered for a purpose other than to prove the truth of the matter asserted.” *Id.* at 3. Petitioner further asserts that the description of the experiments and data in the expert report and letter, which Petitioner refers to as “Dr. Currie’s submission,” is admissible because it is “a more reliable comparator” than the data Patent Owner offers as evidence of unexpected results. *Id.* at 4–5 (citing FRE 807).

Regarding Dr. Peterson’s reliance on such, Petitioner argues that “the Federal Rules of Evidence expressly permit an expert to rely on hearsay in forming an opinion (FRE 703)” and both sides’ experts “have testified on hearsay sources.” *Id.* at 6 (citing Exs. 2027 and 2028). Petitioner also opposes Patent Owner’s argument for exclusion under 37 C.F.R. § 42.65(b), urging that Patent Owner has not moved to strike and “here no data was generated by Dr. Peterson, Mylan, or anyone affiliated with them, and Dr. Currie provided sufficient detail for the purposes for which Dr. Peterson relies upon EX1067.” *Id.* at 14–15.

Patent Owner has access to the complete file history. Paper 61 (“Pet. Opp.”) 6–7.

In reply, Patent Owner argues that “Exhibit 1067 is not a public record,” but rather “an untrustworthy” excerpt of the European Patent Office (EPO) file history. Paper 64 (“PO MTE Reply”) 1. Patent Owner further asserts that any use of Exhibit 1067 for non-hearsay purposes “does not justify its hearsay use” and “Petitioner does not dispute that it offered Exhibit 1067 for the truth of Currie’s testing and results.” *Id.* Thus, Patent Owner urges that “[a]t minimum pages 88-98 and 120-127 of Exhibit 1067 . . . should be excluded.” *Id.* at 1–2.

Having considered the parties’ arguments, Patent Owner’s motion to exclude is granted in part. We exclude pages 88–98 and 120–127 of Exhibit 1067 and paragraphs 153–157, 160, 163, 164, and 166 of Exhibit 1063. The remainder of Patent Owner’s motion is dismissed as moot.²³ Our reasoning is explained below.

We agree with Patent Owner that at least portions of Exhibit 1067 that Petitioner and Dr. Peterson rely upon are inadmissible hearsay. The excerpt of the European Patent (EP) file history in Exhibit 1067 is more than 300 pages long. Petitioner and Dr. Peterson cite a handful of those pages, which come from documents falling into one of two categories: (1) letter briefs and an unsigned “Annex A: experimental report” submitted by counsel for the opponent in an EPO opposition proceeding more than a decade ago (Ex. 1067, 88–98, 120–127), and (2) letter briefs from the EP patentee’s counsel at Mintz Levin in the same proceeding (Ex. 1067, 84–86, 99–103,

²³ Because these materials are excluded we have not considered them in our merits analysis, except where otherwise indicated.

138–140).²⁴ From this first category, Petitioner and Dr. Peterson rely on the descriptions of experiments and data that were apparently carried out by, or at the direction of, the opponent’s witness, Dr. Currie. *See* Reply 23, 26, 28; Ex. 1063 ¶¶ 153–157, 160, 163–164, 166 (citing Ex. 1067, 90–94 120–122, 125–126).²⁵ As does Petitioner, we refer to this first category of materials as “the Currie submission.”

Petitioner relies on the Currie submission for the truth of the matters asserted therein. The Currie submission argues that plecanatide does not exhibit unexpected or improved results over human uroguanylin. *See generally* Ex. 1067, 88–98, 120–127. Petitioner makes essentially the same arguments, urging there is “no difference in activity, heat stability, or topoisomeric interconversion” between uroguanylin and plecanatide, citing the Currie submission as evidence. Reply 23; *see also id.* at 26 (citing Ex. 1067 to urge “additional experiments [that] demonstrate SP-304 does **not** improve GC-C receptor affinity 8-fold versus uroguanylin”), 28 (citing Ex. 1067 to assert “Dr. Currie . . . used routine HPLC optimization to distinguish the SP-304 topoisoforms”). Petitioner also relies on the testimony

²⁴ Exhibit 1067 identifies the EP patentee as “Synergy Pharmaceuticals, Inc.” *See* Ex. 1067, 41. Patent Owner appears to acknowledge that it is related to the EP patentee. *See* Sur-reply 20 (referring to statements in the EP patentee’s letter briefs (i.e., Ex. 1067, 102, 140) as coming from Patent Owner).

²⁵ The documents Petitioner cites explain that Dr. Currie “is the Chief Scientific Officer of Ironwood Pharmaceuticals, Inc., the company that performed the above-mentioned experiments that are reported in Annex A, on behalf of Opponent 1” and request “permission for Dr Currie to make oral submissions at the Oral Proceedings.” Ex. 1067, 126. It is not clear whether Dr. Currie provided an oral submission at any proceeding before the EPO.

of Dr. Peterson who Petitioner says reviewed the Currie submission and “concluded Dr. Currie’s experiments are more reliable than” Patent Owner’s own data. Reply 23 (citing Ex. 1063 ¶¶ 152–161, 163–167). In other words, Petitioner relies on statements (e.g., data) in the Currie submission, which were not offered as testimony in this proceeding, as proof that the results they describe are true. Those statements are, therefore, hearsay. FRE 801(c).

Petitioner’s argument that Exhibit 1067 is admissible as a public record under FRE 803(8) is unavailing. See Pet. Opp. 1–2. Federal Rule of Evidence 803(8) provides an exception to the rule against hearsay for “[a] record or statement of a public office” that “sets out: (i) the office’s activities; (ii) a matter observed while under a legal duty to report . . . ; or (iii) in a civil case . . . findings from a legally authorized investigation.” FRE 803(8). The only portions of Exhibit 1067 that Petitioner relies upon, i.e., the letter briefs and an expert report submitted by counsel—not the EPO—are none of those things. See *Jordan v. Binns*, 712 F.3d 1123, 1132 (7th Cir. 2013) (explaining the three categories of public records under FRE 803(8)).

In addition, the public record exception does not apply because Patent Owner has sufficiently shown “that the source of information” in the Currie submission and the circumstances of its submission “indicate a lack of trustworthiness.” FRE 803(8)(B). “The public-records exception is justified on the assumption that public officials will perform their duties properly and *without bias*.” *Jordan*, 712 F.3d at 1132 (emphasis added). But it cannot be assumed that the Currie submission was prepared without bias because it is the legal briefing of the patent opponent describing experiments conducted specifically for the purpose of the opposition proceeding. See Ex. 1067, 90 (stating that the “Opponent 1 has asked Ironwood Pharmaceuticals, Inc. to

perform experiments” to respond to the data the EP patentee presented in the opposition proceeding). The Currie submission contains no sworn testimony, nor does the record suggest that Dr. Currie, or anyone else with personal knowledge of the tests described in the Currie submission, has ever testified or been subject to cross-examination regarding the same.²⁶ The mere fact that these materials were submitted to the EPO and as a result entered into the EP file history does not make the statements in the Currie submission any more trustworthy. Thus, we see no reason why that would qualify the hearsay in the Currie submission to be excepted under FRE 803(8) to the general rule against hearsay.

Petitioner’s assertion that “Exhibit 1067 is admissible for the same reason USPTO prosecution histories are admissible” is incorrect. Pet. Opp. 2. To be sure, there are instances when a file history qualifies under the FRE 803(8) exception. *See Valve Corp. v. Ironburg Inventions Ltd.*, 8 F.4th 1364, 1374 (Fed. Cir. 2021) (explaining that file histories offered to authenticate a reference and show the Office had found that it was published before the critical date are admissible because “[t]he patent examiner’s determination of the publication date is a ‘factual finding[] from a legally authorized investigation’” and “‘is supported by sufficient guarantees of trustworthiness’”) (quoting FRE 803(8)(A)(iii) and FRE 807(a)(1)). But the Currie submission is not a factual finding by the EPO. Nor is there is anything in the record suggesting that the EPO ever made any finding based on the experiments and data in the Currie submission. To the contrary, both

²⁶ Patent Owner’s counsel represents that Patent Owner did not have an opportunity to cross-examine Dr. Currie in the EPO proceeding. Tr. 50:14–17.

sides represent that the EPO did not credit or otherwise make findings based on the data in the Currie submission. *See* Tr. 25:12–18 (Petitioner’s counsel stating that the EPO “found itself unable to conclude as to credit any of the data”); 50:1–13 (Patent Owner’s counsel stating that “the Currie data was not credited” and that the EPO made no findings based upon it); *see also* Ex. 2069, 104:15–19 (Dr. Peterson testifying that he understood that the EPO did not credit the data).

It is also true, as Patent Owner points out (*see* PO MTE Reply 1), that our rules expressly address the admissibility of USPTO records. *See, e.g.*, 37 C.F.R. § 42.61(b) (“Certification is not necessary as a condition to admissibility when the evidence to be submitted is a record of the Office to which all parties have access.”). But even for a United States patent or application, our rules state that a “specification or drawing . . . is admissible as evidence only to prove what the specification or drawing describes.” *Id.* at § 42.61(c). “If there is data in the specification or a drawing upon which a party intends to rely to prove the truth of the data, an affidavit by an individual having first-hand knowledge of how the data was generated must be filed.”²⁷ *Id.*; *see also* 37 C.F.R. § 42.65(b) (requiring an affidavit where “a party relies on a technical test or data from such a test”). Such rules reflect a desire for information to evaluate, potentially through cross-examination, the source of the data regarding the specifics of the tests performed to obtain it. Again, Petitioner has not provided anything like that for the Currie submission, leaving open questions as its trustworthiness that prevent application of the FRE 803(8) exception.

²⁷ For example, Patent Owner has provided such an affidavit here for the data it relies upon from the Specification. *See* Ex. 2066 ¶¶ 2, 5–9.

Petitioner’s reliance on the residual exception to hearsay under FRE 807 is similarly unavailing. *See* Pet. Opp. 5–6. The residual exception requires “sufficient guarantees of trustworthiness—after considering the totality of circumstances under which it was made and evidence, if any, corroborating the statement.” FRE 807(a)(1). For the reasons explained above, such guarantees are not present here. The residual exception also requires the statement be “more probative on the point for which it is offered than *any other evidence that the proponent can obtain* through reasonable efforts.” FRE 807(a)(2) (emphasis added). Petitioner offers the Currie submission as evidence of experiments and data that it contends rebut Patent Owner’s unexpected results showing. But Petitioner has not demonstrated that it could not have reasonably conducted its own experiments to try to obtain rebuttal data.

The non-hearsay uses Petitioner alleges for some of the other parts of Exhibit 1067 do not justify admission of the hearsay in the Currie submission. Citing the letter briefs of the EP patentee’s counsel in Exhibit 1067, Petitioner urges that Patent Owner “repeatedly asserted to the EPO [that] it was ‘impossible’ to purify uroguanylin.” Pet. Opp. 3 (quoting Ex. 1067, 140); *see also* Reply 23 (citing Ex. 1067, 102–103, 139–140).²⁸ According to Petitioner, the EP patentee’s statements in these letter briefs are not hearsay because they are admissions of a party opponent under FRE 801(d)(2) or they are admissible for impeachment or similar reasons. Pet.

²⁸ Petitioner incorrectly asserts that the EPO found “‘contradiction’ in [Patent Owner’s] positions about purifying uroguanylin.” Pet. Opp. 3 (quoting Ex. 1067, 59). The statement Petitioner cites appears in a section of the EPO decision (14.2.4.1.1) summarizing the opponent’s argument; it is not the EPO’s finding. *See* Ex. 1067, 588–89.

Opp. at 4 (citing FRE 803(14)–(15), 804(b)(3) and 806). But even so, Petitioner’s argument plainly does not apply to the Currie submission, which is not a statement or admission from Patent Owner, but rather statements from an adverse party. Accordingly, we agree with Patent Owner that the Currie submission (i.e., pages 88–98 and 120–127 of Exhibit 1067) is hearsay without exception and therefore should be excluded under FRE 802.

The fact that Petitioner has not provided an affidavit under 37 C.F.R. § 42.65(b) for the Currie submission is a separate basis for its exclusion. There is no affidavit from Dr. Currie in the record. While Petitioner argues that Dr. Peterson disclosed “all the facts and data on which his opinions are based,” Petitioner does not contend that his testimony somehow meets the affidavit requirements under § 42.65(b). *See* Pet. Opp. 14–15. Indeed, Dr. Peterson testified that he had never performed a T84 cell assay himself and had not communicated with Dr. Currie regarding these tests. *See* Ex. 2069, 10:15–20, 15:22–16:2. Thus, his testimony is not a substitute for the missing affidavit. Indeed, Patent Owner raises several questions regarding “how the Currie tests were performed and how the Currie data were generated” that could affect its veracity, but that remain unexplained because an affidavit containing the information required under 37 C.F.R. § 42.65(b) was not offered. *See* PO MTE 4. We see no reason to excuse Petitioner from complying with the rule simply because it and Dr. Peterson are not “affiliated” with Dr. Currie. Pet. Opp. at 15. Similarly, Petitioner’s argument that we should excuse its non-compliance because Patent Owner has moved to exclude rather than strike is, on the record before us here, form over substance. *See id.* at 14.

We disagree with Petitioner’s argument that excluding the Currie submission on this basis would also require us to exclude the Liu reference on which Patent Owner relies because a “declaration from Liu’s authors” has not been provided. *Id.* at 14. Liu is a peer-reviewed journal article published prior to and independent of the EPO opposition that describes tests and results conducted for research purposes,²⁹ whereas the Currie submission describes tests from an expert report conducted for the purpose of a legal proceeding. *See* Ex. 1067, 90. The former is not within the scope of the rule. *See* 37 C.F.R. § 42.65 (describing requirements for “Expert testimony; tests and data”). Thus, pages 88–98 and 120–127 of Exhibit 1067 are separately excluded because Petitioner has not provided an affidavit under 37 C.F.R. § 42.65(b) for the testing described therein.

We also grant Patent Owner’s motion to exclude the portions of Dr. Peterson’s second declaration relying upon the tests and data in the Currie submission. It is well established that data relied upon by an expert need not itself be admissible in order for the expert opinion to be admissible, “[i]f experts in the particular field would reasonably rely on those kinds of facts or data in forming an opinion.” FRE 703; *Wi-LAN Inc. v. Sharp Elecs. Corp.*, 992 F.3d 1366, 1375–76 (Fed. Cir. 2021). “The burden to establish reasonable reliance is on the proponent of challenged expert testimony.” *Wi-*

²⁹ Liu was published in the journal *Anticancer Research* in 2009 and therefore predates the start of the EPO opposition proceeding. Ex. 2046, 3777; Ex. 1067, 44 (stating that Opponent 1 filed its notice of opposition on April 22, 2010). None of the authors listed on Liu are ostensibly affiliated with Patent Owner, nor does Liu suggest that their work was supported by Patent Owner. *See* Ex. 2046, 3777 (listing authors and their affiliations), 3782 (acknowledgements). Moreover, Petitioner does not challenge Patent Owner’s description of Liu as an “independent” study. *See* Resp. 66.

LAN, 992 F.3d at 1376. Here, Petitioner has not shown that experts in the field typically rely on sources like the Currie submission, i.e., legal briefing describing experiments conducted solely for the purpose of an EPO opposition proceeding without any accompanying testimony.

The *Wi-LAN* case is informative. There, the Federal Circuit affirmed the district court’s decision excluding unauthenticated source code printouts³⁰ and the testimony of the proponent’s expert regarding that source code. *See Wi-LAN*, 992 F.3d at 1370–76. The court explained that it “is obviously correct” that “experts typically rely on material, like source code,” but the proponent had “not made a showing that source code experts reasonably rely on unauthenticated source code printouts.” *Id.* at 1376. The court also agreed with the district court that FRE 703 should not be used as a “backdoor to allow the admission of otherwise inadmissible . . . materials.” *Id.* at 1375 (internal quotations omitted).

In this case, it is clear that experts in the present field typically rely on experiments and test data such as T84 cell assays and HPLC chromatographs. However, Petitioner has not shown that they do so when the only source of such experiments and data is legal briefing generated for litigation purposes without any accompanying testimony or opportunity to cross-examine of someone with personal knowledge of the experiments and data. Nor do we find it appropriate under the circumstances here to permit

³⁰ The proponent in *Wi-LAN* provided “source code printouts” from third-party chip manufacturers “together with declarations from employees of the manufacturers purporting to authenticate the source code printouts.” *Id.* at 1371. However, the proponent “did not establish that the declarants would be available to testify at trial” and as a result they could not be used to authenticate the source code. *See id.* at 1372–74.

Petitioner to use FRE 703 and Dr. Peterson's testimony as a backdoor to admit otherwise inadmissible test data, notwithstanding the rules requiring an affidavit to support such. Therefore, because the Currie submission is inadmissible hearsay and Petitioner has not met its burden to show that FRE 703 applies, we grant Patent Owner's motion to exclude Dr. Peterson's testimony describing and opining regarding the experiments and data in the Currie submission (i.e., Ex. 1063 ¶¶ 153–157, 160, 163, 164, and 166). We also exclude those paragraphs because, as explained above, Petitioner has not provided an affidavit with the information required by required by 37 C.F.R. § 42.65(b) for the tests and data Dr. Peterson relies upon from the Currie submission.

Finally, we turn to the portions of Exhibit 1067 Petitioner relies upon that are not part of the Currie submission, i.e., the EP patentee's statements in the letter briefs from Mintz Levin regarding the alleged impossibility of purifying uroguanylin. *See* Ex. 1067, 103 (referring to tests “using uroguanylin in its naturally-occurring form (i.e a mixture of A- and B-forms)” which is “clinically relevant, as [the data] was obtained with the peptides in the form in which they are therapeutically administered”), 140 (“because of the rapid isomeric interconversion of uroguanylin (and the much slower interconversion of E³-uroguanylin (SP-304)), it is impossible to compare the affinity of the active conformation (only) of uroguanylin to the affinity of the active confirmation of [SP-304]”). Petitioner cites these statements for its argument that Patent Owner failed to test what Petitioner contends is the closest prior art, i.e., isomerically pure uroguanylin containing only the fully active topoisomer. It is true that a party or declarant's prior inconsistent statements may be admissible notwithstanding

the rule against hearsay. *See, e.g.*, FRE 801(d)(2), 806. However, we agree with Patent Owner that the EP patentee’s statements in Exhibit 1067 are not inconsistent with the position it takes here i.e., that it is “not possible to compare plecanatide with *only* the active conformation of human uroguanylin” because “[s]ome amount of interconversion to the inactive conformation was inevitable” and therefore the appropriate comparison is between SP-304 and uroguanylin as it naturally occurs. *See* Sur-reply 20. Moreover, as explained above, Patent Owner has shown that its unexpected results evidence compares plecanatide to the closest prior art. *Supra* § III.E.iii. Thus, even considering the EP patentee’s statements that Petitioner points to in Exhibit 1067, our findings and conclusion as to obviousness remain the same. For this reason, we dismiss as moot Patent Owner’s motion to exclude the remainder of Exhibit 1067. We also dismiss as moot Patent Owner’s motion to exclude paragraphs 34, 149, 152, 162 and 167 of Exhibit 1063, which rely only on the patentee’s statements in Exhibit 1067 and not the Currie submission or the data therein.

V. PETITIONER’S MOTION TO EXCLUDE

Petitioner moves to exclude four categories of exhibits: (1) Exhibits 2001–2007, (2) Exhibits 2024 and 2025, (3) Exhibits 2027 and 2028, and (4) Exhibit 2040. Pet. MTE 1. Patent Owner opposes Petitioner’s motion. Paper 59 (“PO Opp.”). We address each category below.

A. Exhibits 2001–2007

Petitioner argues these exhibits should be excluded under FRE 402 and 403 because they were offered only “to support a pre-institution argument—*Fintiv*—that was not preserved during trial.” Pet. MTE 1. We do

not rely on any of these exhibits in rendering our decision on Petitioner's grounds. We, therefore, dismiss as moot Petitioner's motion to exclude Exhibits 2001–2007.

B. Exhibits 2024 and 2025

Petitioner moves to exclude the declarations of Patent Owner's experts, Dr. Davies (Ex. 2024) and Dr. Waldman (Ex. 2025). Pet. MTE 1. Patent Owner argues Dr. Davies and Dr. Waldman applied a legally erroneous standard for their testimony regarding the lead compound analysis and reasonable expectation of success. *See id.* at 2–5. Petitioner also objects to their testimony regarding Figure 3 of Li, the pKa of “buried amino acid residues . . . and amino acid prevalence,” and “human uroguanylin [being] prone to undesirable levels of topoisomeric interconversion.” *See id.* at 5–8. According to Petitioner, Dr. Davies's and Dr. Waldman's testimony on each of these subjects “should be excluded for being unreliable and lacking basis.” *Id.* Finally, Petitioner contends their testimony relying on Exhibits 2027, 2028, and 2040 should be excluded because those exhibits are not admissible. *Id.* at 8.

Patent Owner contends, “Petitioner's Motion improperly argues the weight of the evidence, rather than its admissibility, and is further deficient because Petitioner mischaracterizes the relevance of the challenged Exhibits and manufactures reliability concerns where none exist.” PO Opp. 1.

We agree with Patent Owner that Petitioner's concerns regarding the testimony of Dr. Davies and Dr. Waldman go to the weight to be accorded such testimony and are not a sound basis for excluding it from the record. *See* PTAB Consolidated Trial Practice Guide (Nov. 2019) 79 (“A motion to

exclude must explain why the evidence is not admissible (e.g., relevance or hearsay) but may not be used to challenge the sufficiency of the evidence to prove a particular fact.”). Our analysis of the merits addresses each of the issues Petitioner points to in its motion and gives appropriate weight to Exhibits 2024 and 2025 in view of the record as whole. *See supra* § III.E. Moreover, as explained below, Exhibits 2027 and 2028 are admissible and we do not rely on Exhibit 2040. For these reasons, we are not persuaded by Petitioner’s arguments and therefore deny its motion to exclude Exhibits 2024 and 2025.

C. Exhibits 2027 and 2028

Petitioner moves to exclude Exhibits 2027 and 2028 under FRE 403, FRE 802 and FRE 901. Pet. MTE 9. According to Petitioner, these documents “purport to be reports created in 2015, more than a decade after the claimed critical date of the invention, [and] signed only by the first named inventor, Kunwar Shailubhai,” which “describe studies conducted in late 2001 . . . based on the work of ‘Dr. Surendra Dheer,’ who has not provided any testimony in this proceeding.” *Id.* at 9 (quoting Ex. 2027, 1; Ex. 2028, 1). Petitioner urges that “[f]ar from being contemporaneous, well-attested corroborating evidence, these exhibits are belated, self-serving, and unauthenticated derivations purportedly of someone else’s unpublished notebooks.” *Id.* at 10. Petitioner also asserts that “[i]f these exhibits are admitted, their scope should be restricted to minimize prejudice so far as possible.” *Id.*

We do not agree that Exhibits 2027 and 2028 should be excluded. As an initial matter, Petitioner’s authenticity argument is unpersuasive.

Documents are authenticated by evidence “sufficient to support a finding that the item is what the proponent claims it is.” FRE 901(a). Patent Owner has provided that evidence in the form of testimony from Dr. Shailubhai. Dr. Shailubhai is the named author of the reports in these exhibits. Ex. 2027, TRUL00018203; Ex. 2028, TRUL00018260. Dr. Shailubhai testifies that Exhibits 2027 and 2028 are true and correct copies of these reports and that he has first-hand knowledge of both the reports and the test data disclosed therein. Ex. 2066 ¶¶ 5, 10, 17, 30, 32–34. That testimony is sufficient to authenticate Exhibits 2027 and 2028.

Dr. Shailubhai’s testimony is also sufficient to establish the reports in Exhibits 2027 and 2028 as business records under the FRE 803(6) hearsay exception. *See* Ex. 2066 ¶ 31 (testifying regarding the FRE 803(6) requirements). Indeed, Petitioner does not articulate any specific argument to the contrary in its motion. *See* Pet. MTE 9–10.

In its reply, Petitioner asserts that “[d]ocuments created long after the fact, without testimony from the person who conducted the study or corroboration from anyone but an inventor and that have later been modified, can hardly be characterized as ordinary business records.” Paper 65, 4. We disagree for several reasons. First, Patent Owner offered a witness, Dr. Shailubhai, with first-hand knowledge and he testified extensively regarding the specifics of the tests and data. *See generally* Ex. 2066. Indeed, Dr. Shailubhai explained that he directed both the testing in the studies and the preparation of the reports on those studies. *Id.* ¶¶ 10, 17, 32–33. He also approved those reports. *Id.* ¶¶ 32–33. Petitioner cross-examined Dr. Shailubhai (*see* Ex. 1061) and identifies nothing deficient regarding this testimony.

Second, Petitioner’s argument that as an inventor, Dr. Shailubhai, cannot corroborate his own corroborating documents is inapposite. *See* Pet. MTE 9 (citing *Apator Miitors ApS v. Kamstrup A/S*, 887 F.3d 1293, 1296–97 (Fed. Cir. 2018)). Petitioner confuses the corroboration standard for establishing an earlier invention date with the rules for admitting documents into evidence. *See Apator*, 887 F.3d at 1295 (describing the “rule of reason” analysis for assessing whether an inventor’s testimony regarding invention date is sufficiently corroborated by independent evidence). Nothing prevents an inventor from authenticating and establishing as business records data and reports of which they have personal knowledge.

Third, the dates on the study reports do not undermine Dr. Shailubhai’s testimony. Dr. Shailubhai testifies that the records for these studies “were made or obtained at or near the time of occurrence of the matters set forth in the records.” Ex. 2066 ¶ 31. Again, Petitioner cites no contrary testimony in support of its motion. The mere fact that the reports in Exhibits 2027 and 2028 show “Study Dates” in 2001, “Final Report Date[s]” in 2008 and subsequent amendments in 2015 does not contradict Dr. Shailubhai’s testimony. Ex. 2027, TRUL00018203; Ex. 2028, TRUL00018260; *see also* Ex. 2066 ¶¶ 32, 33. Moreover, as we noted above, while the EC₅₀ values in one of those reports (Exhibit 2027) was amended, the underlying data reports remained the same throughout. *Supra* § III.E.iii.2 n.19.

Finally, Petitioner’s argument under FRE 403 is unavailing. The evidence in Exhibits 2027 and 2028 is relevant to Patent Owner’s unexpected results showing. Petitioner’s arguments regarding the reliability of the data Patent Owner relies upon goes to the weight of the evidence, not

its admissibility. For these reasons, we deny Petitioner’s motion to exclude Exhibits 2027 and 2028.

D. Exhibit 2040

Petitioner argues that Exhibit 2040, “a post-filing date (March 31, 2004) letter from Michael Pennington, Ph.D[.], to Dr. Kunwar Shailubhai,” should be excluded under FRE 403, FRE 802, and FRE 901. Pet. MTE 10–11. We do not rely on this exhibit in rendering our decision on Petitioner’s grounds. We, therefore, dismiss as moot Petitioner’s motion to exclude Exhibit 2040.

VI. CONCLUSION

Petitioner has not shown, by a preponderance of the evidence, that claims 1–6 of the ’786 patent are unpatentable.

Claim(s)	35 U.S.C. §	Reference(s)/ Basis	Claim(s) Shown Unpatentable	Claim(s) Not Shown Unpatentable
1	103(a)	Currie, Li		1
2, 4, 5	103(a)	Currie, Li, Narayani		2, 4, 5
3–5	103(a)	Currie, Li, Narayani, Campieri		3–5
6	103(a)	Currie, Li, Ekwuribe		6
Overall Outcome				1–6

Further regarding the parties’ motions to exclude, we grant-in-part Patent Owner’s motion and deny Petitioner’s motion.

VII. ORDER

Accordingly, it is:

ORDERED that Petitioner has not shown that claims 1–6 of U.S. Patent 7,041,786 B2 are unpatentable;

FURTHER ORDERED that Patent Owner's Motion to Exclude (Paper 53) is granted-in part as indicated above;

FURTHER ORDERED that Petitioner's Motion to Exclude (Paper 54) is denied; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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